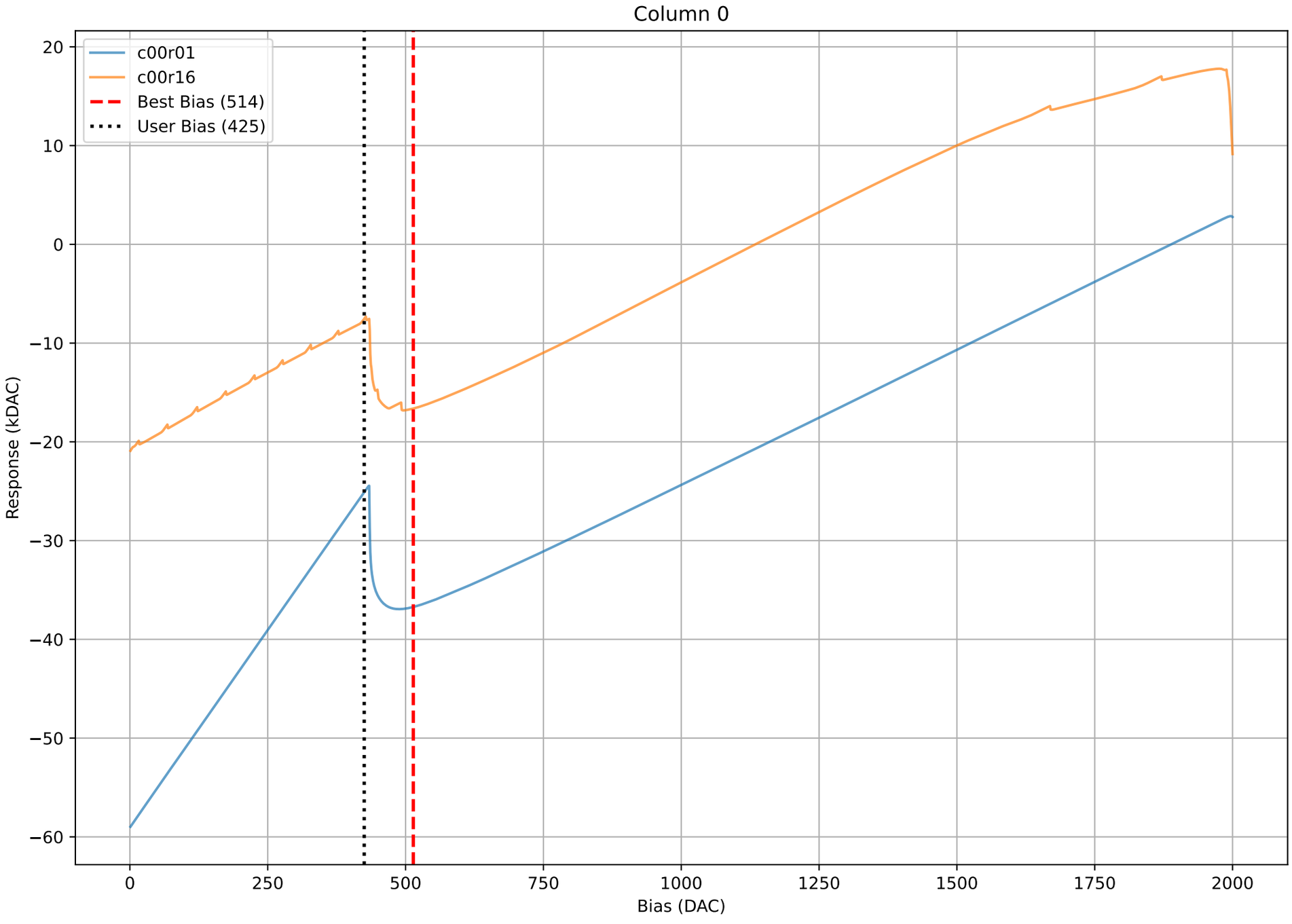
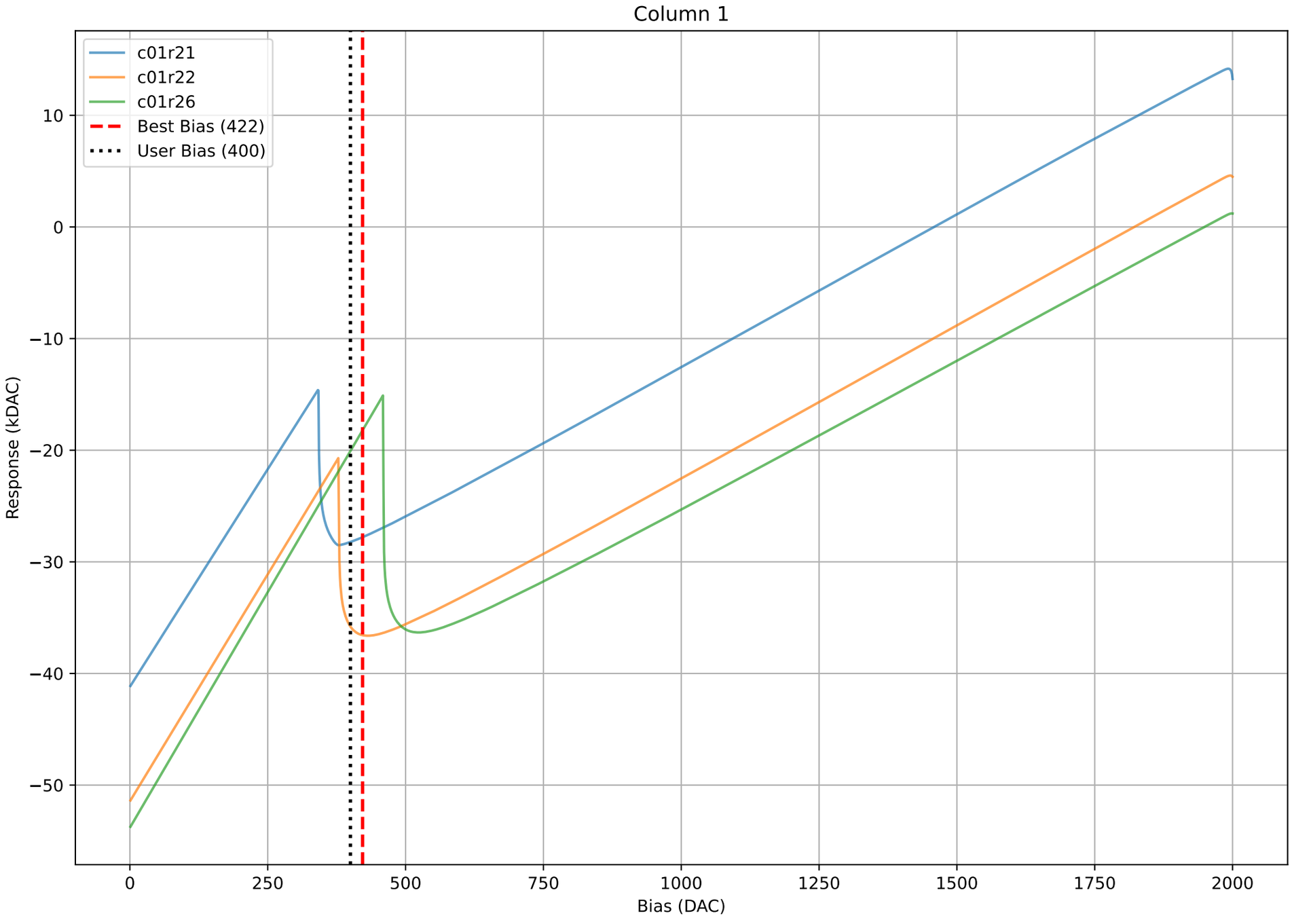
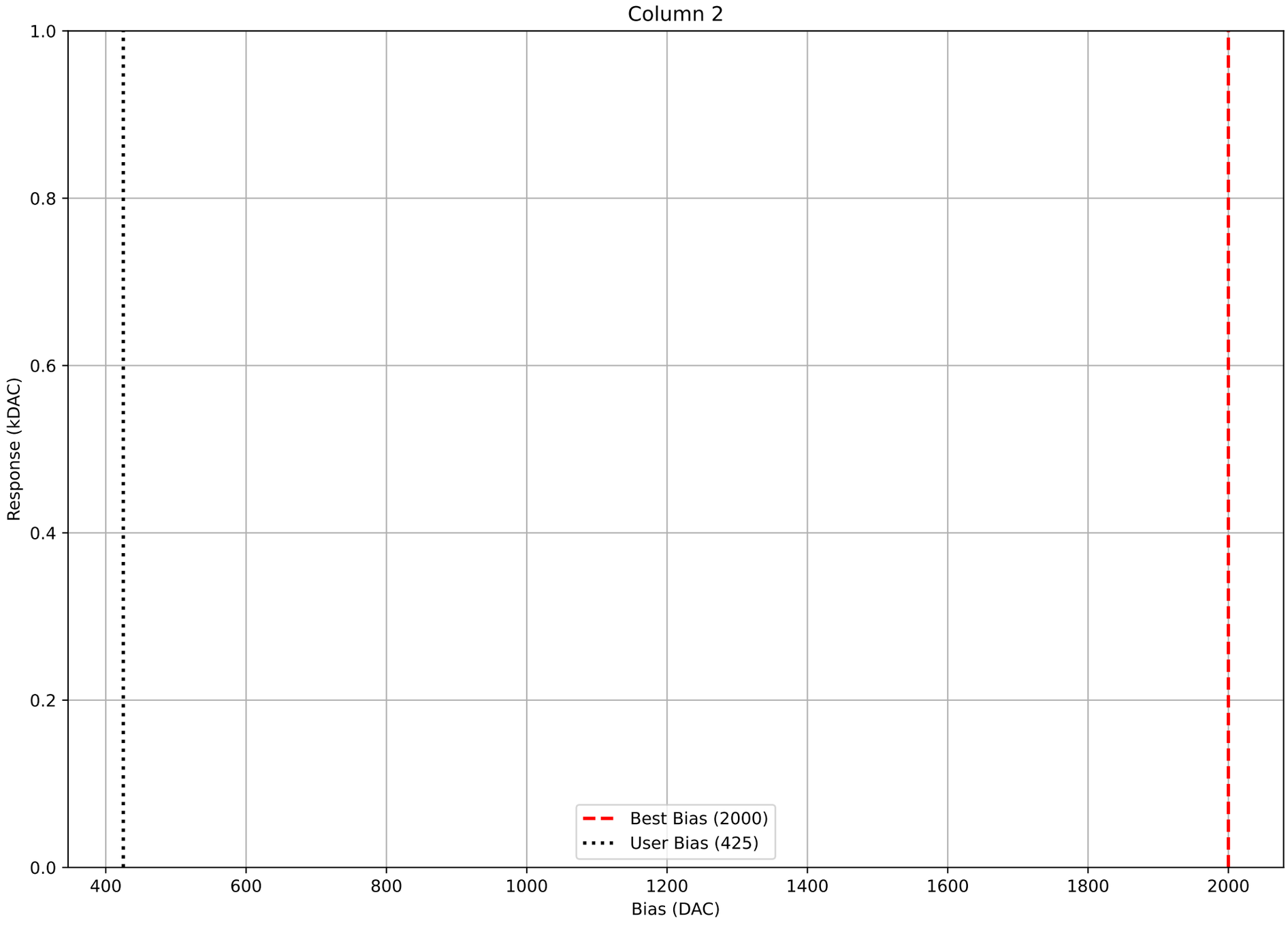
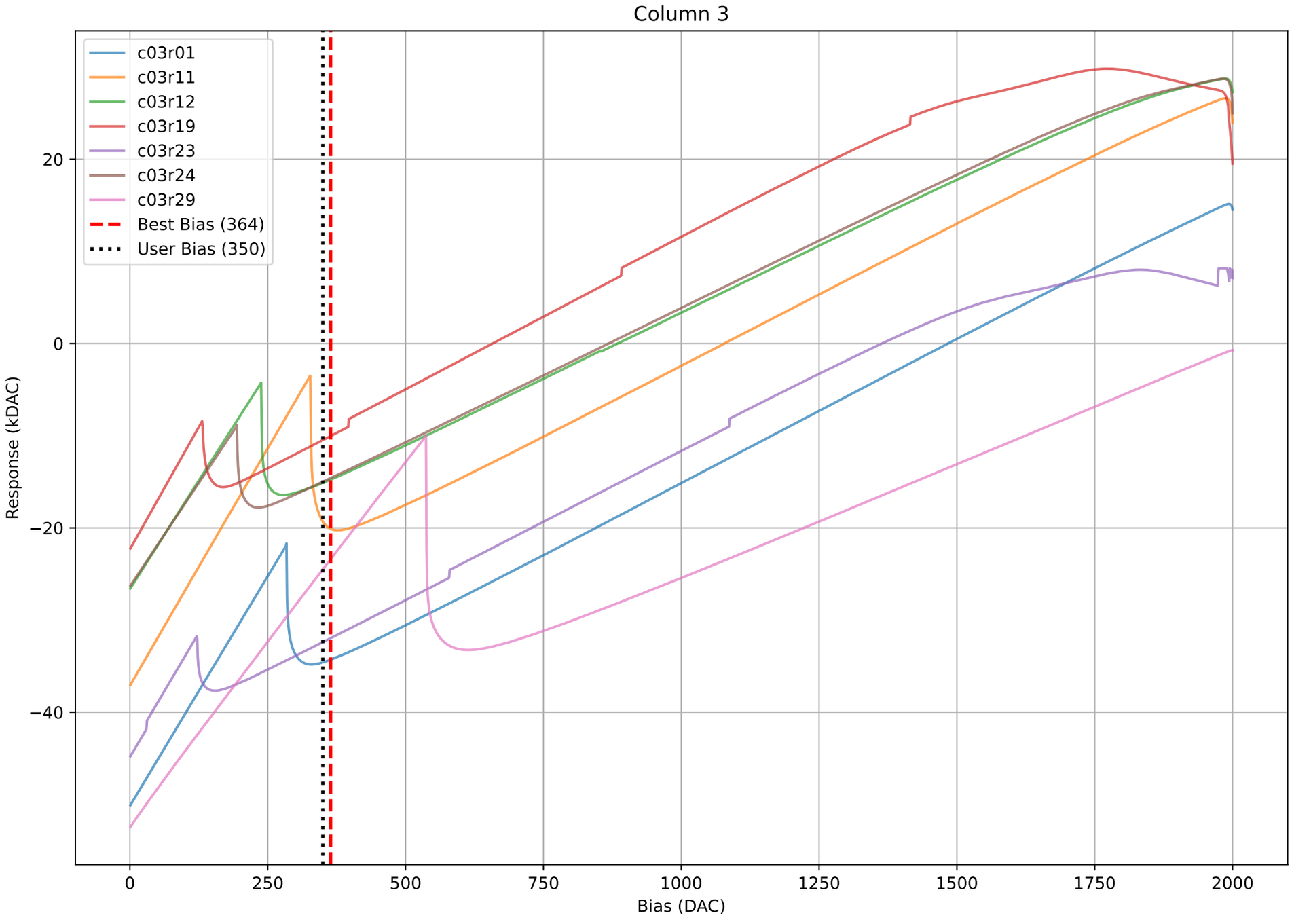
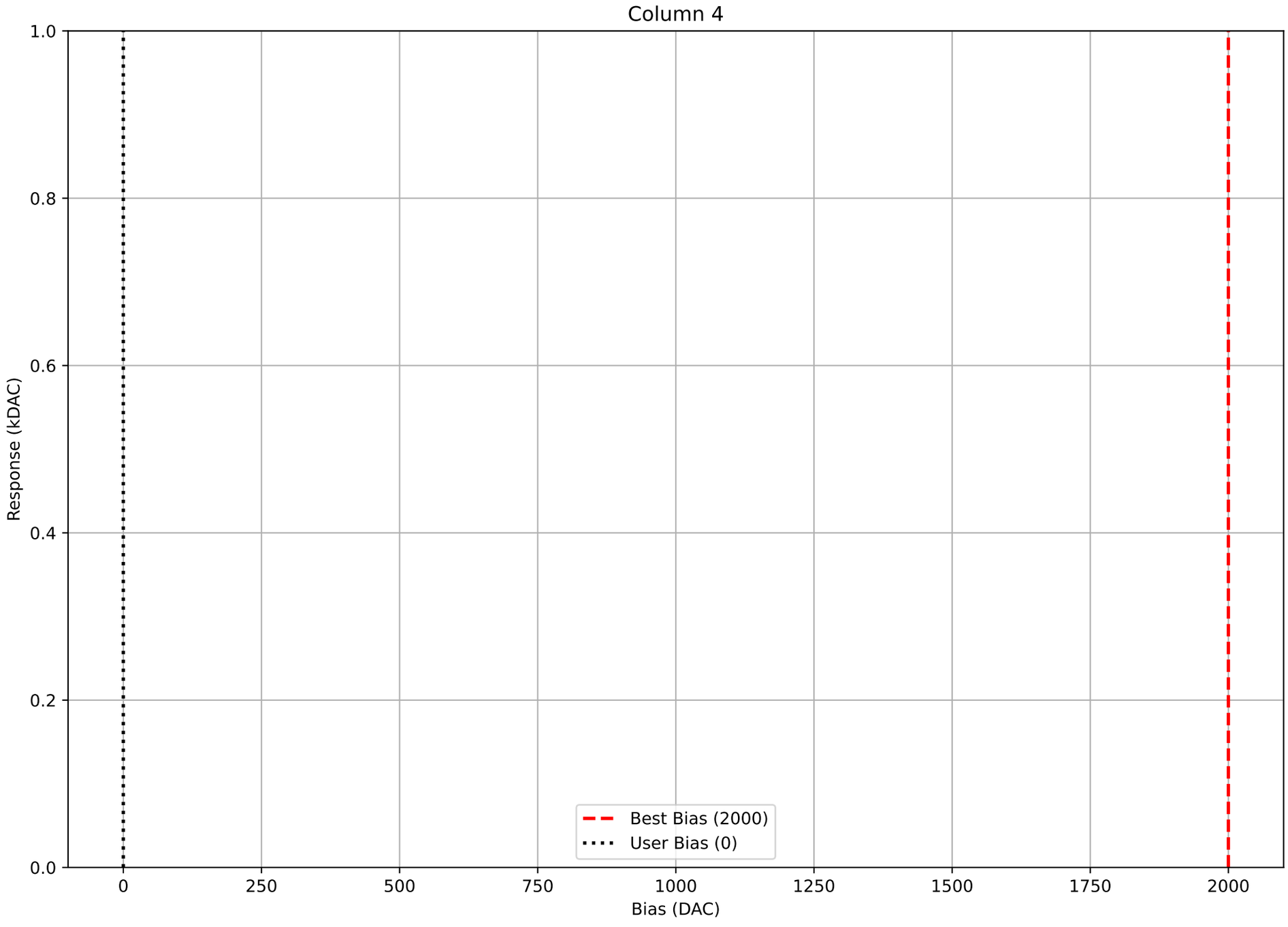
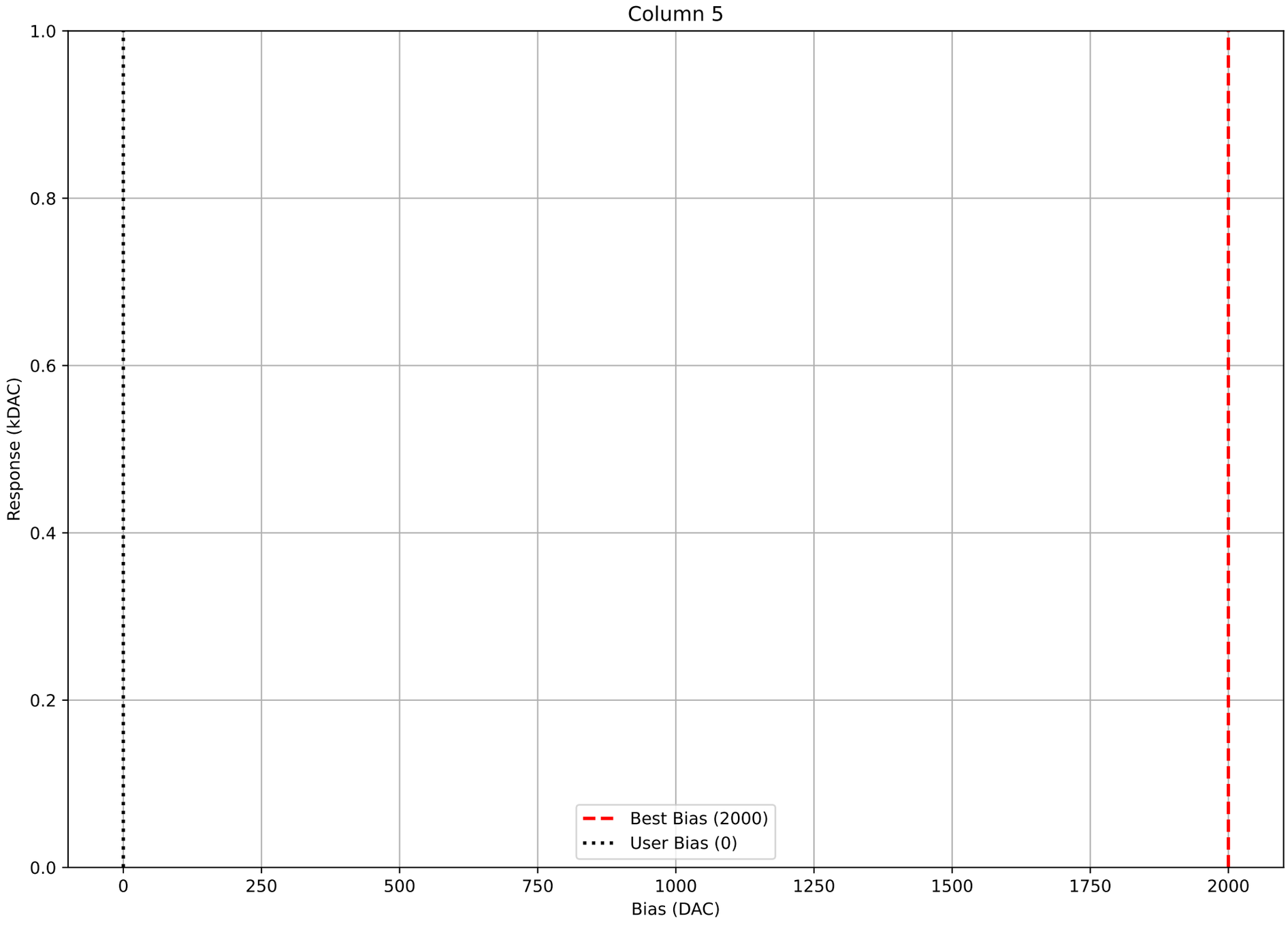

Column 0
c00r01
c00r16
Best Bias (514)
User Bias (425)
Response (kDAC)
Bias (DAC)
20
10
0
−10
−20
−30
−40
−50
−60
0
250
500
750
1000
1250
1500
1750
2000

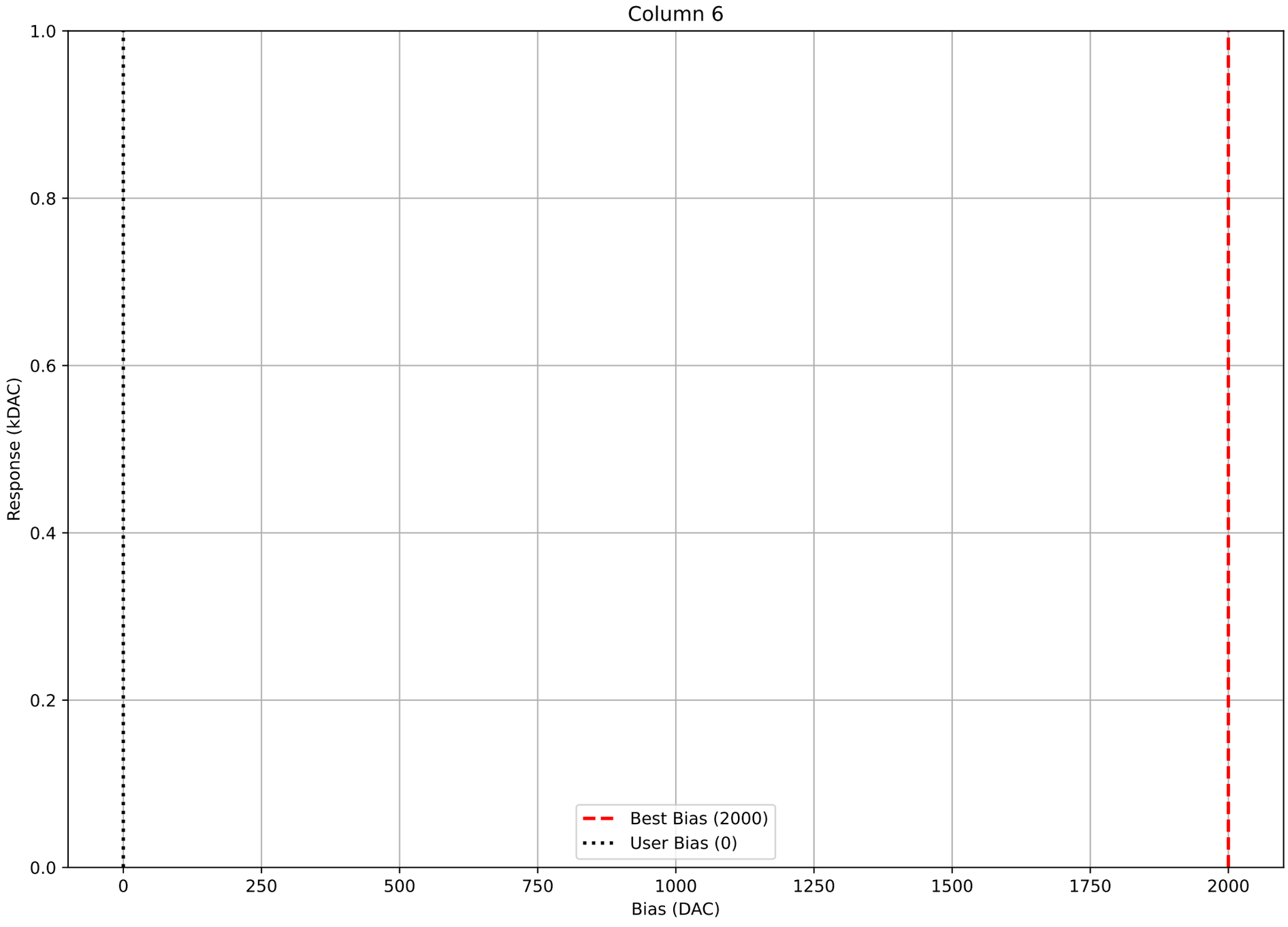

Column 1
c01r21
c01r22
c01r26
Best Bias (422)
User Bias (400)
Response (kDAC)
10
0
−10
−20
−30
−40
−50
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)

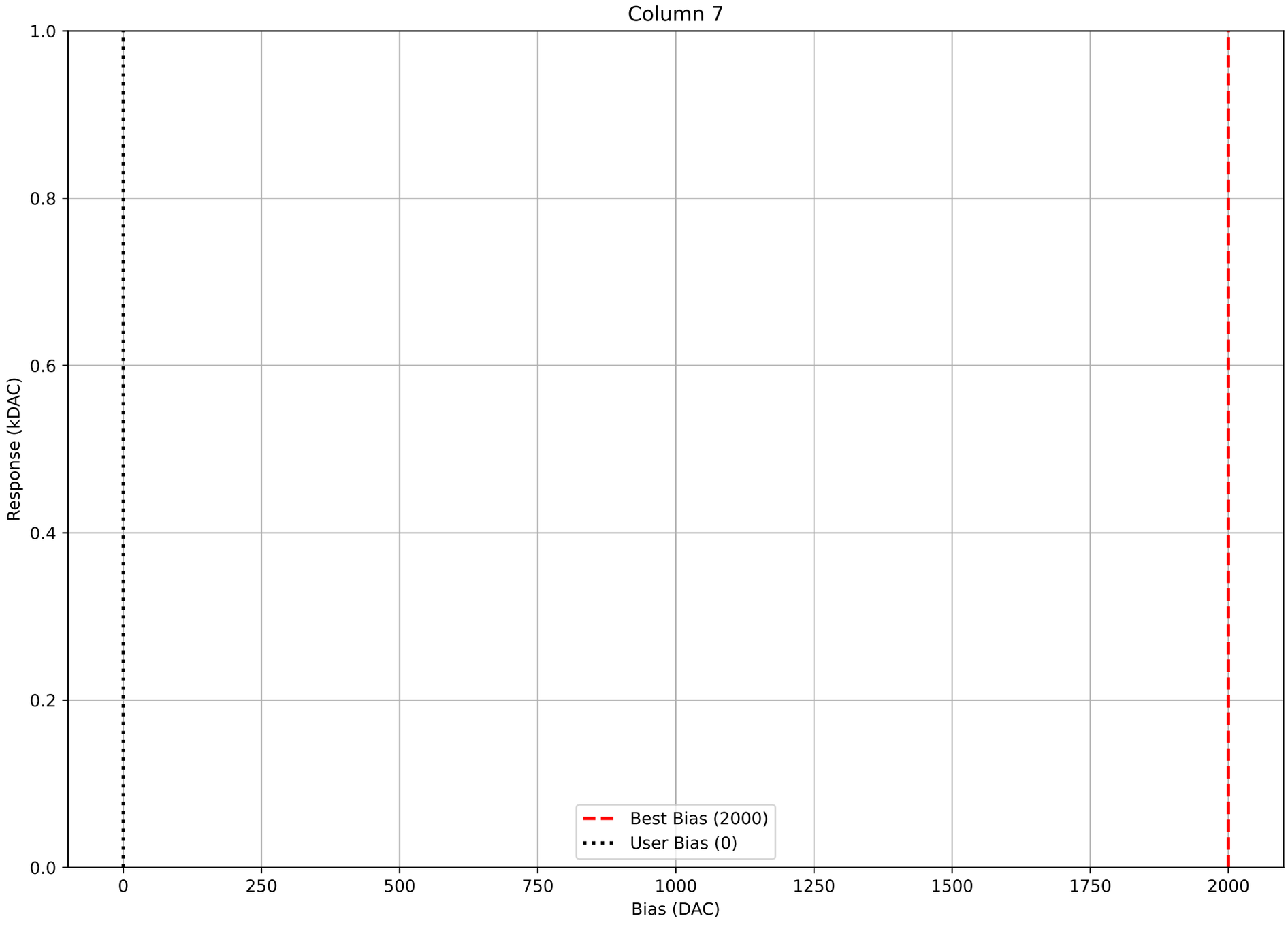

Column 2
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
400
600
800
1000
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (425)

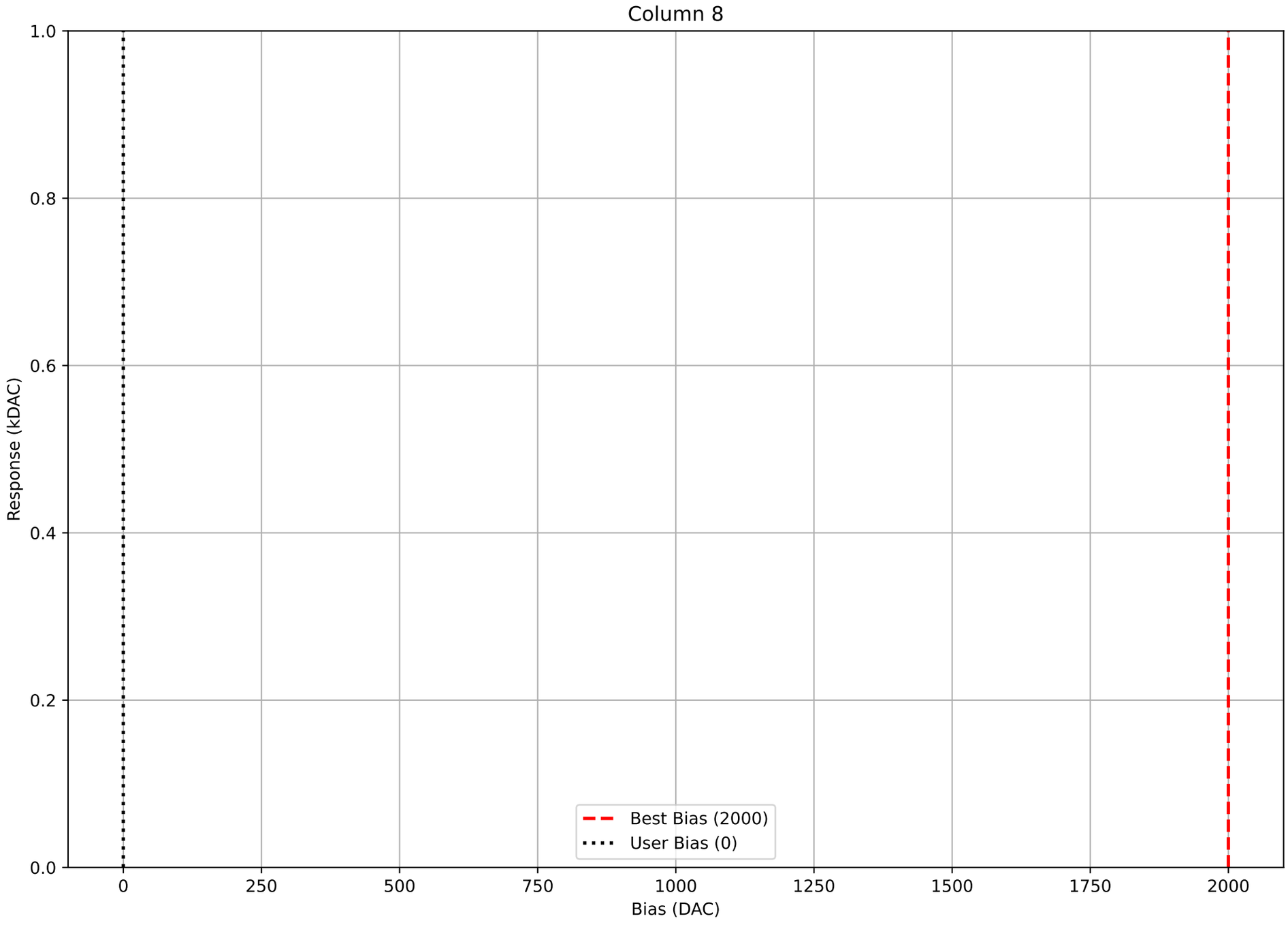

Column 3
c03r01
c03r11
c03r12
c03r19
c03r23
c03r24
c03r29
Best Bias (364)
User Bias (350)
Response (kDAC)
20
0
−20
−40
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)

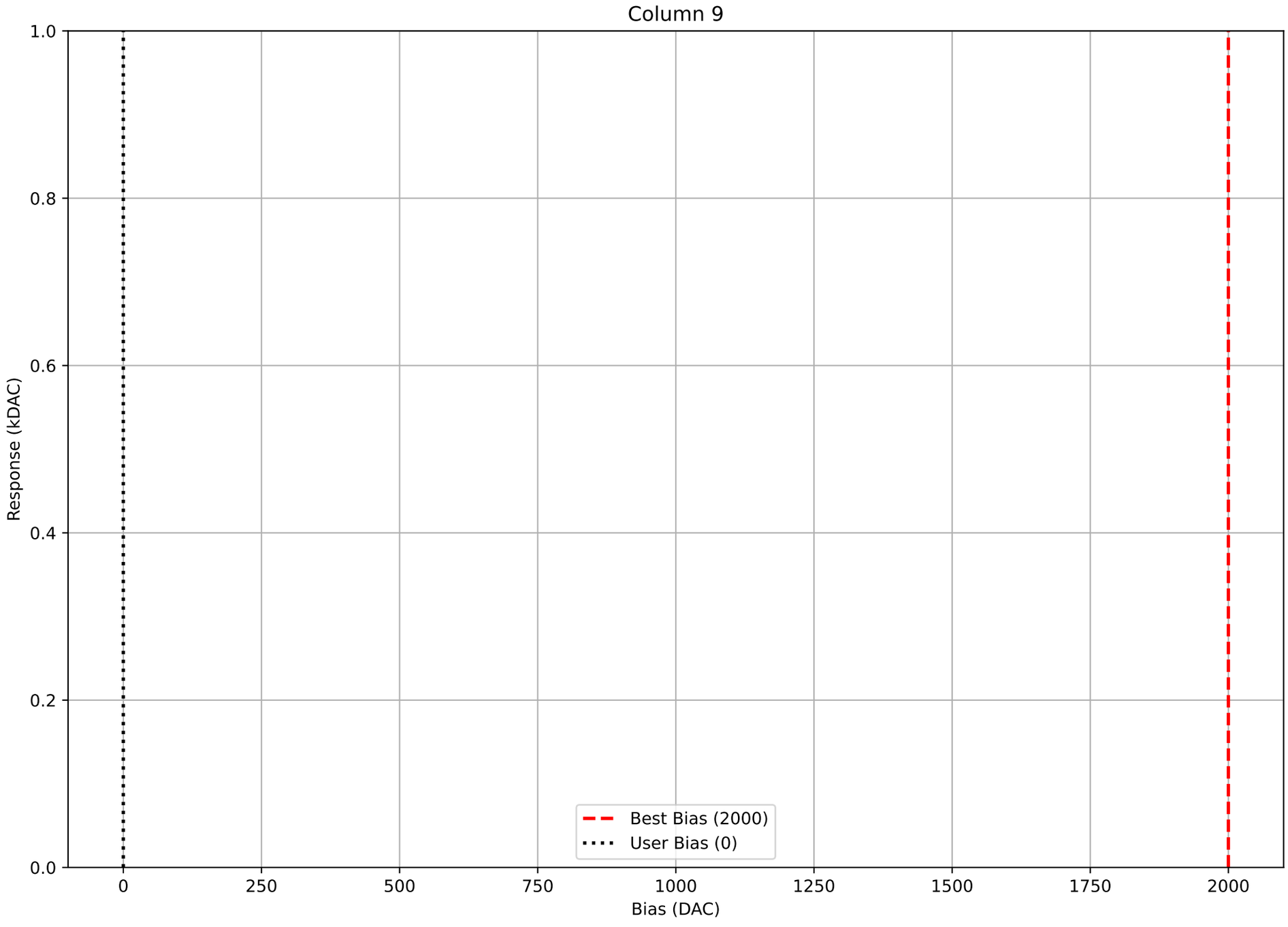

Column 4
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

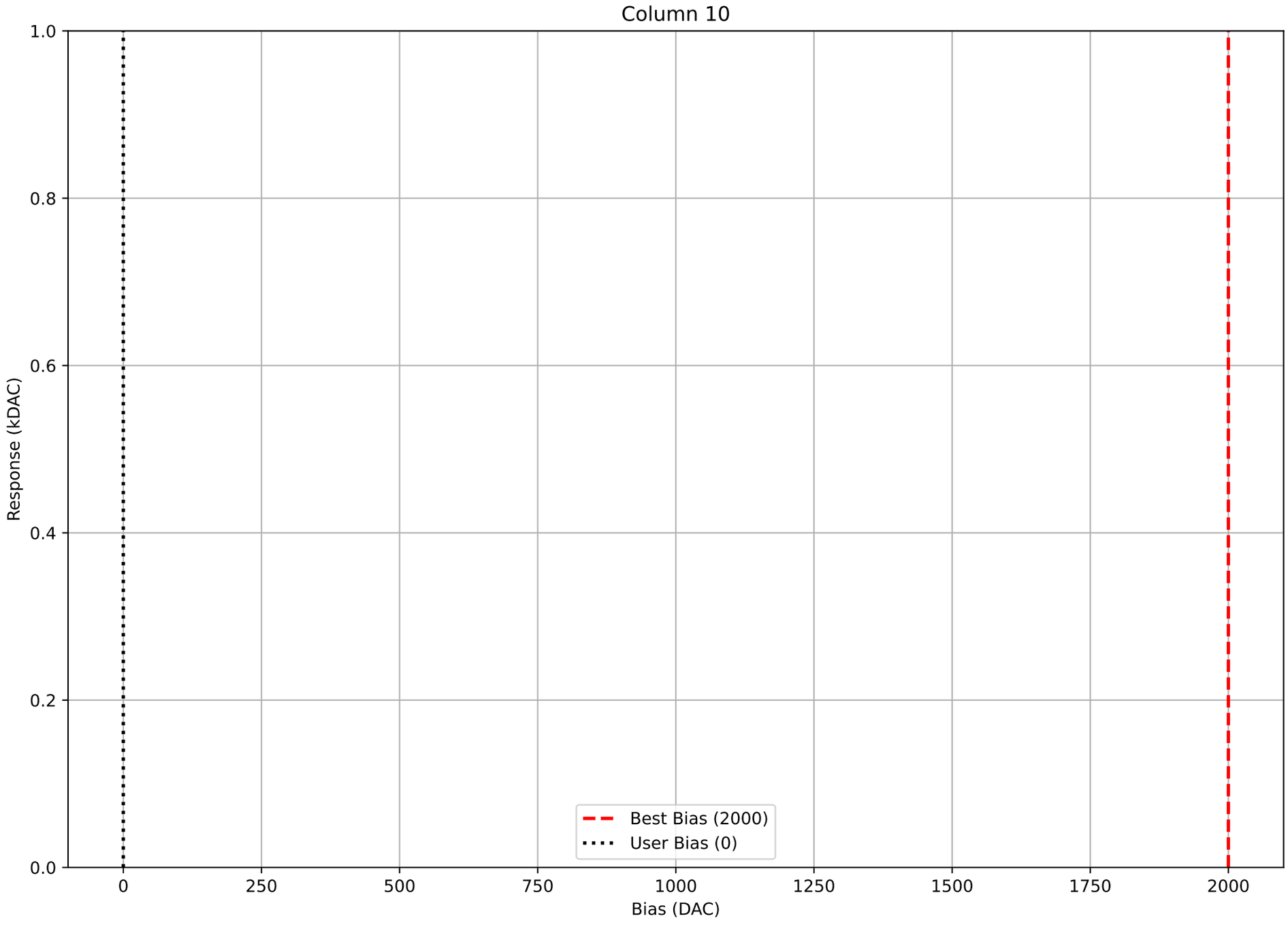

Column 5
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

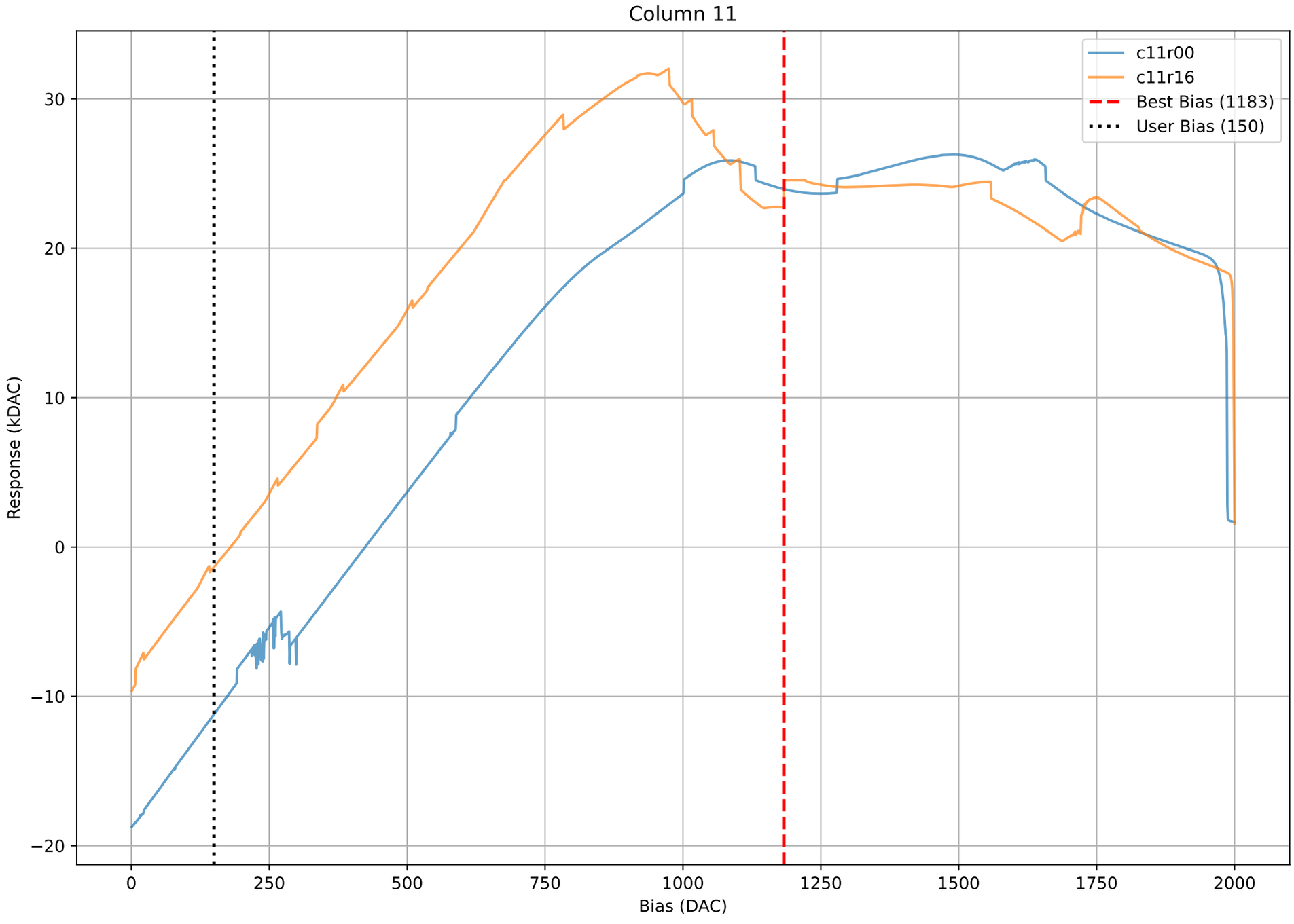

Column 6
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

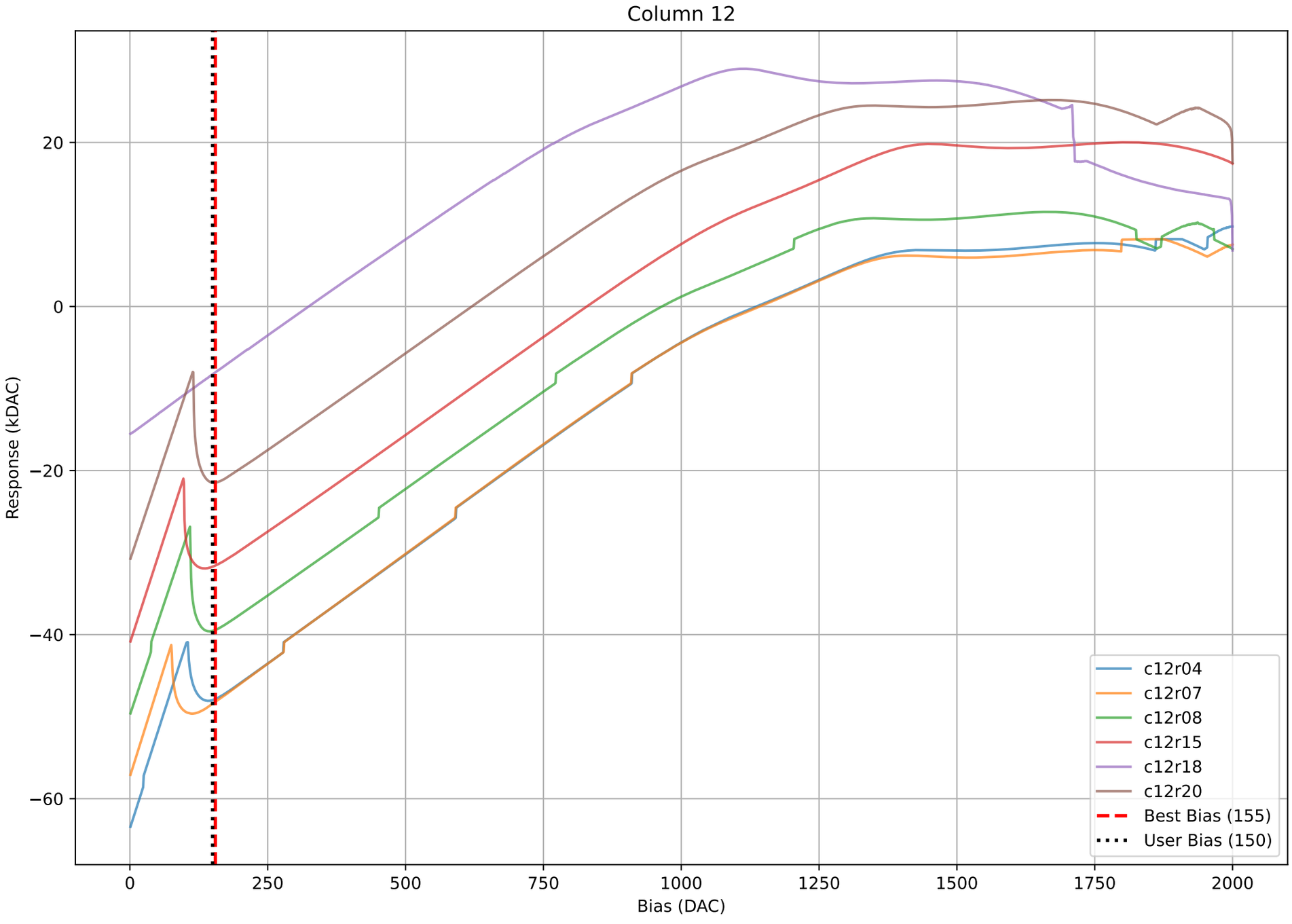

Column 7
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

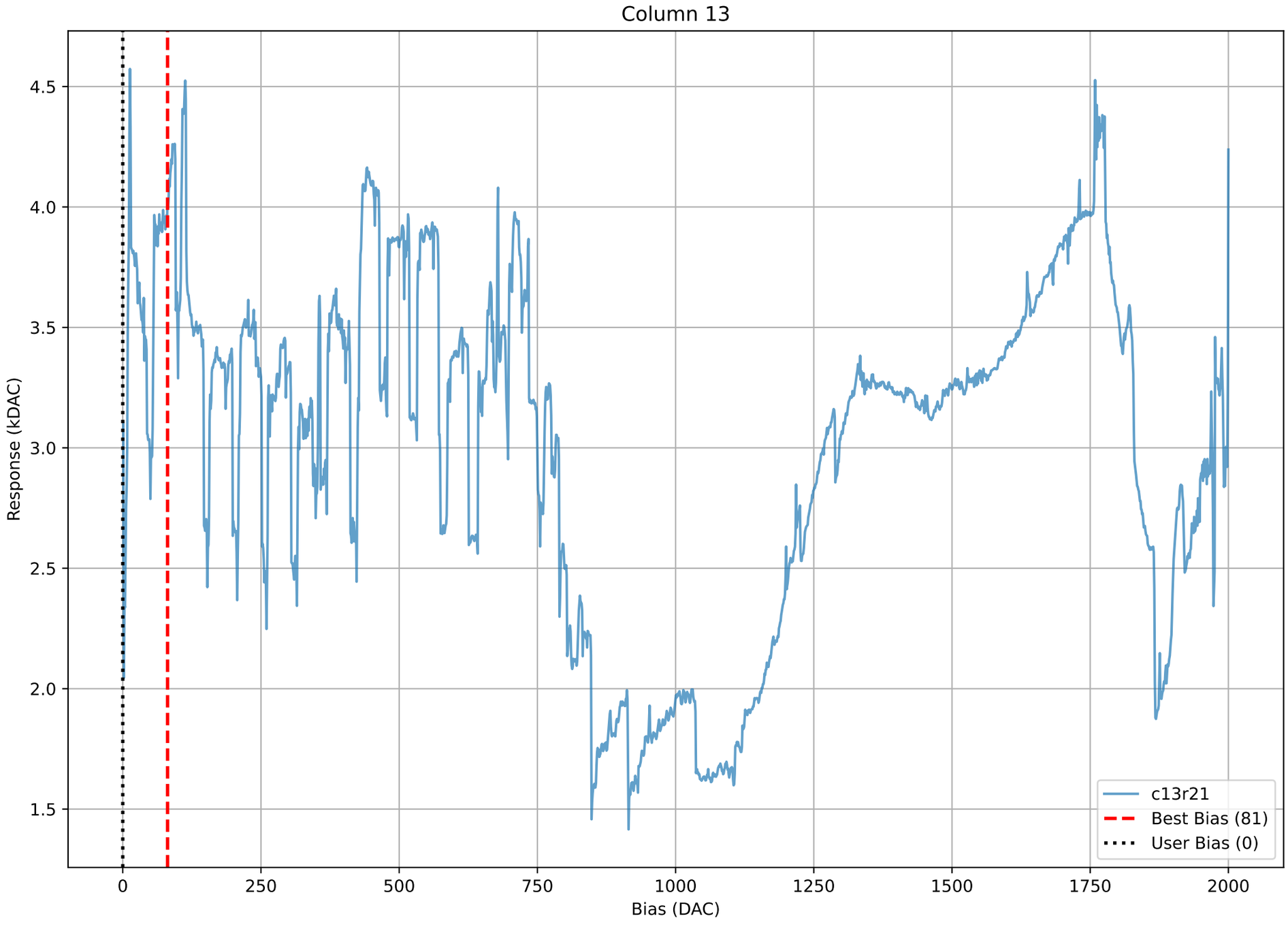

Column 8
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

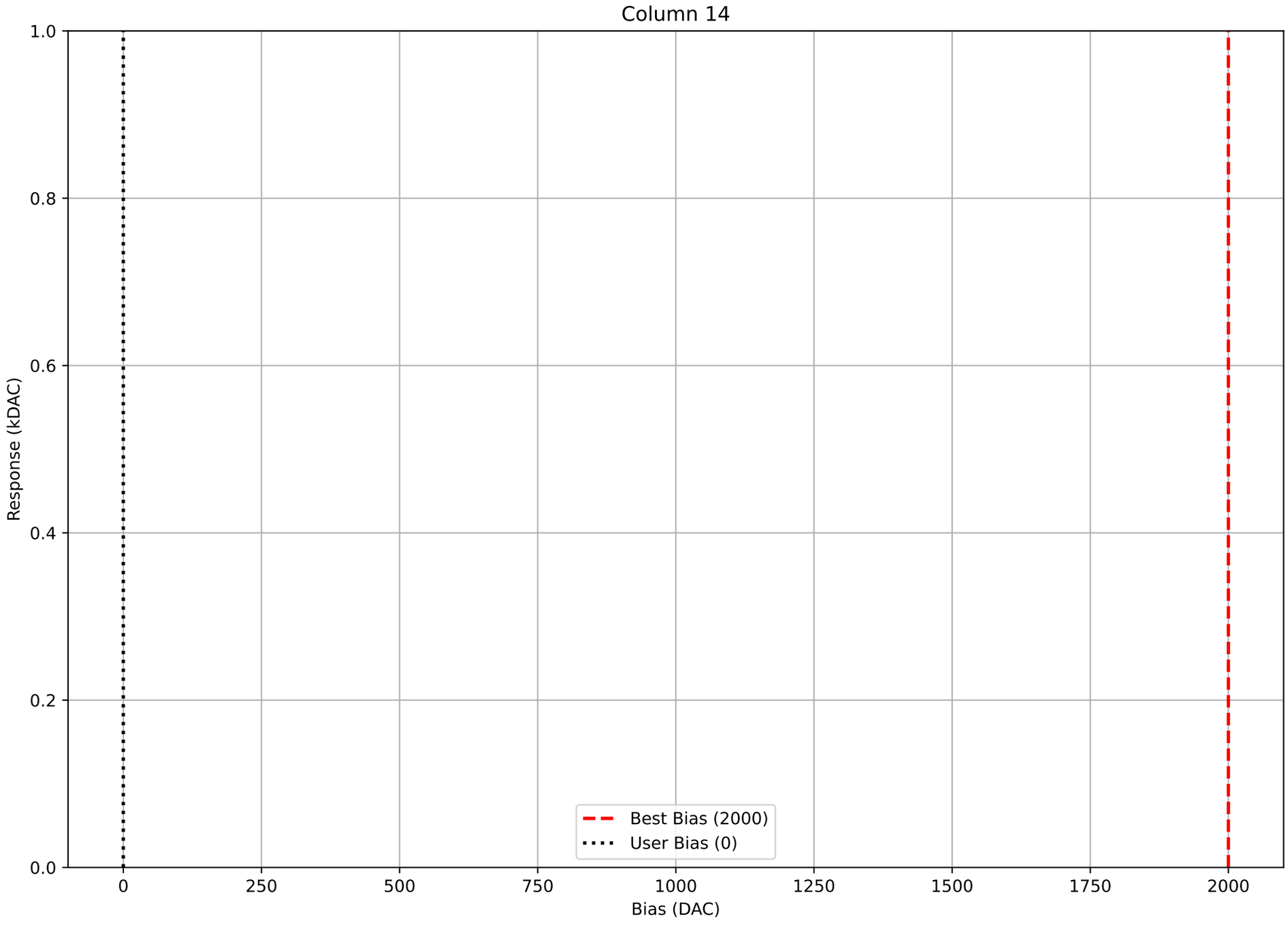

Column 9
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

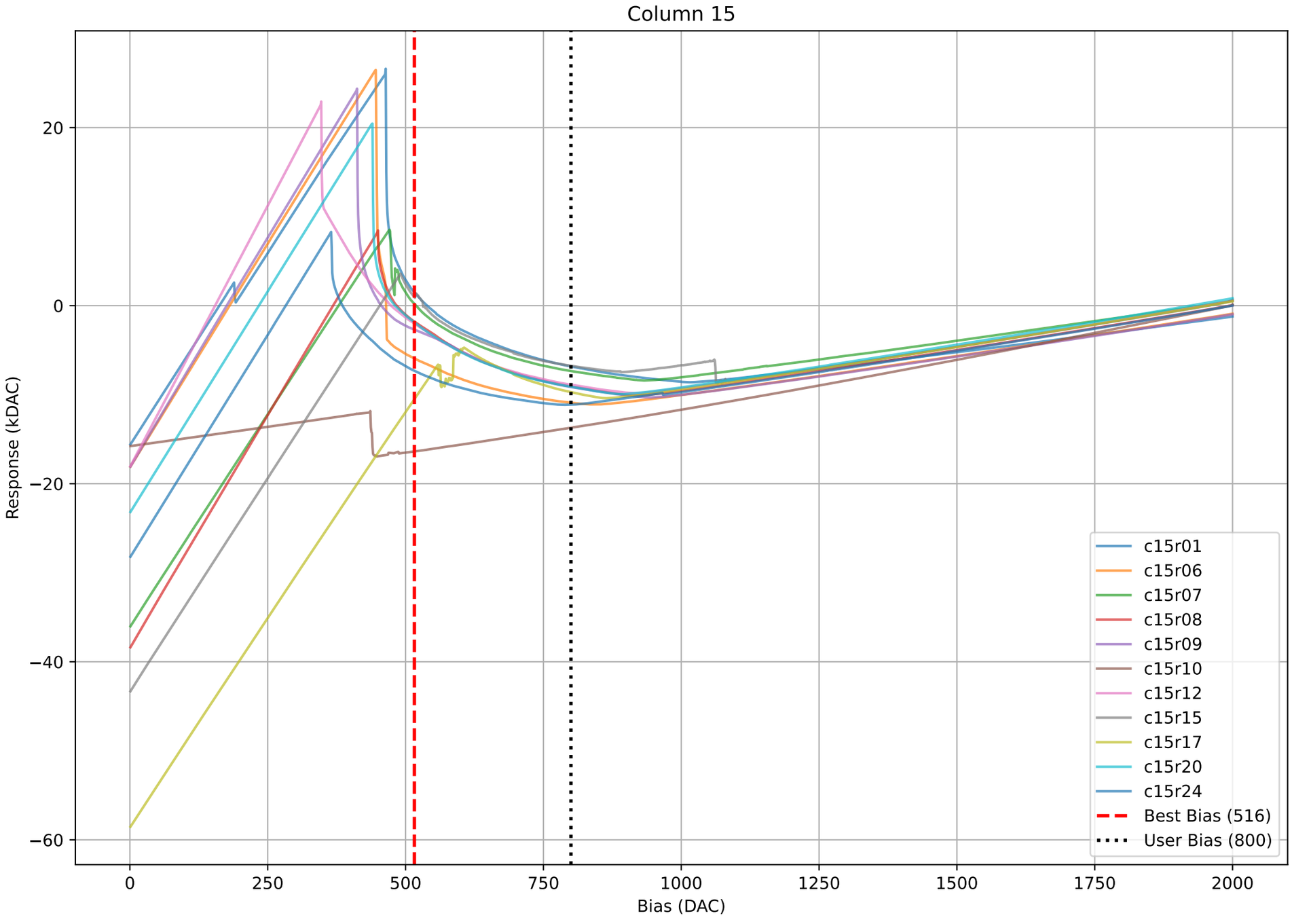

Column 10
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

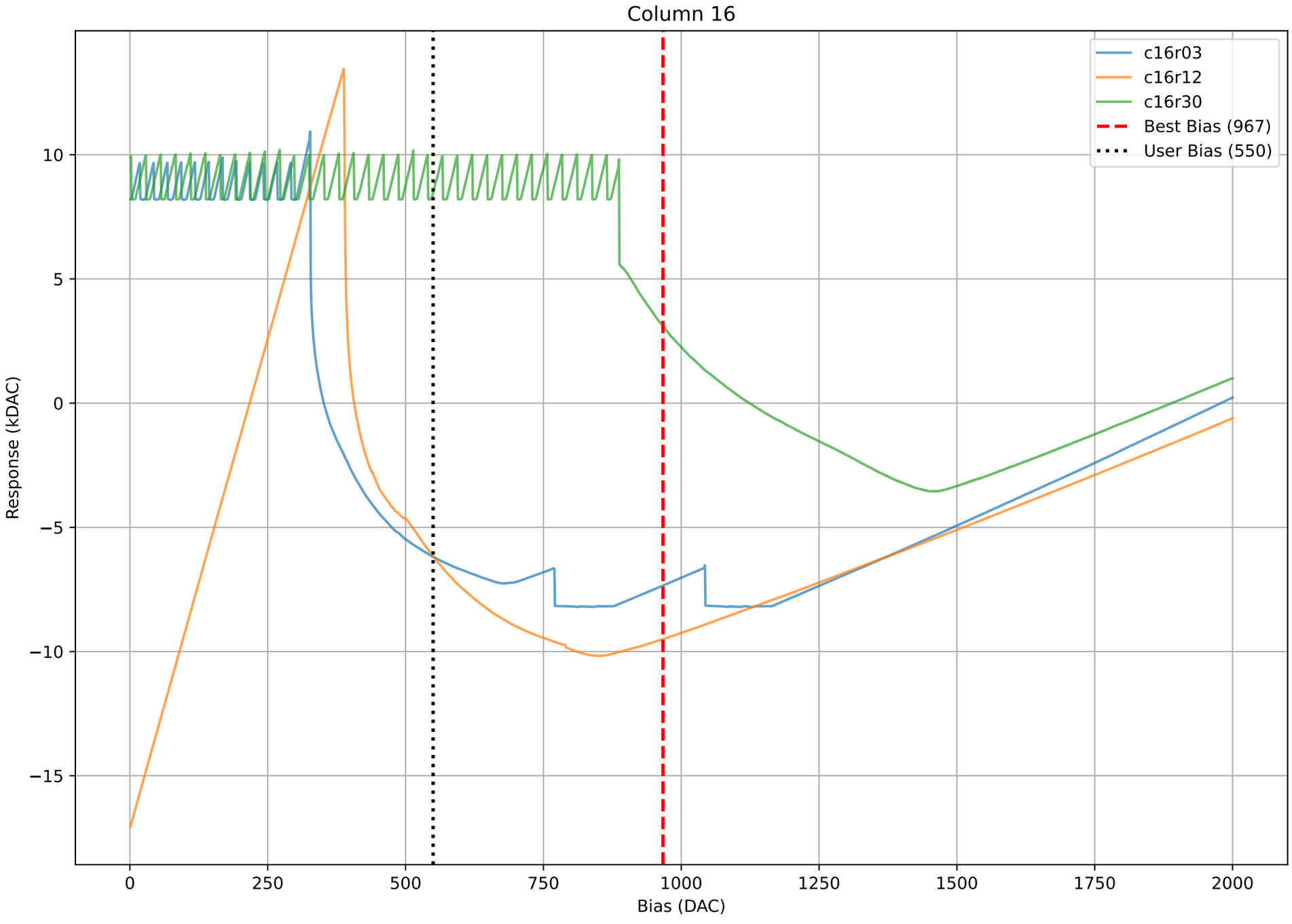

Column 11
c11r00
c11r16
Best Bias (1183)
User Bias (150)
Response (kDAC)
Bias (DAC)
30
20
10
0
−10
−20
0
250
500
750
1000
1250
1500
1750
2000

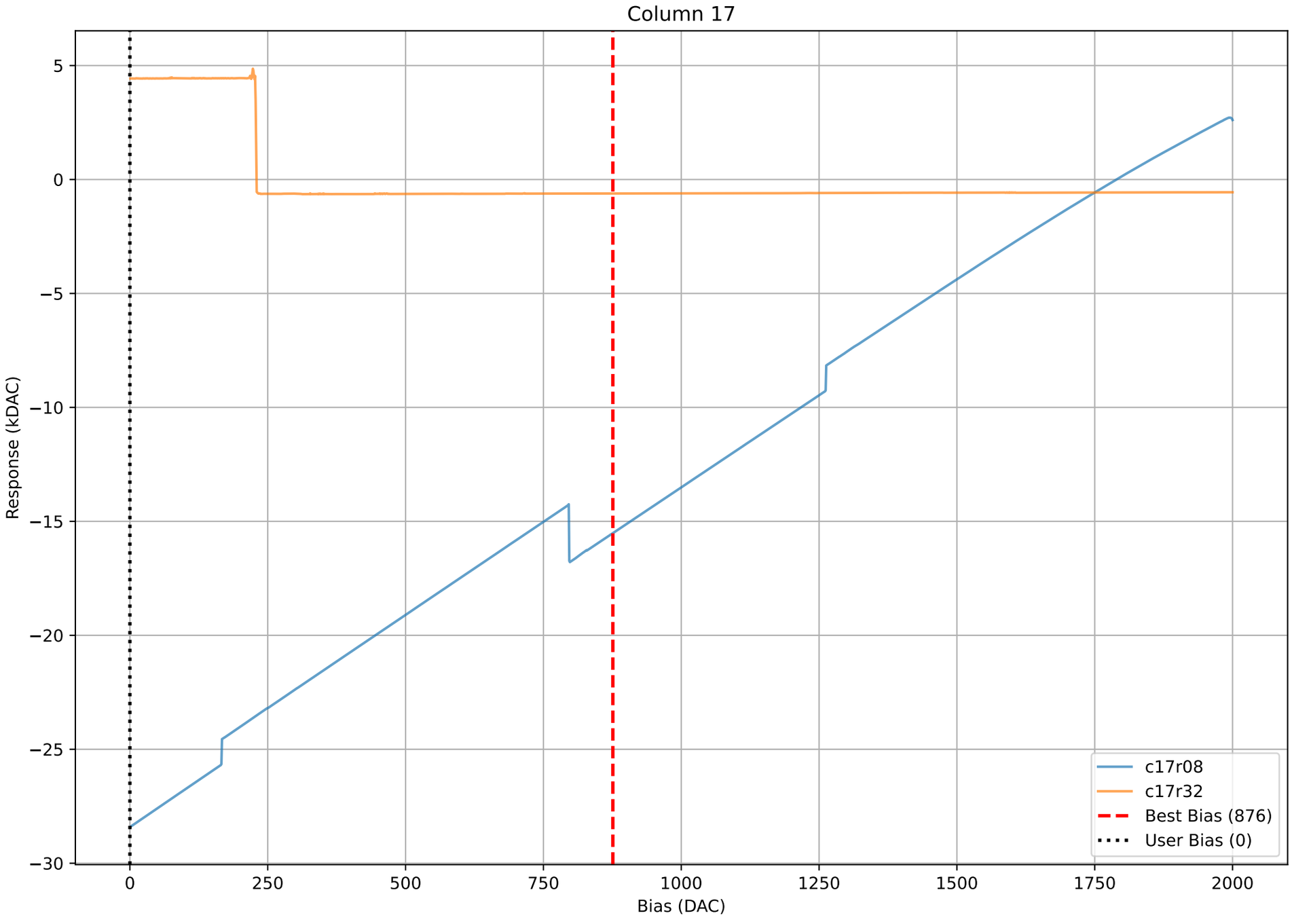

Column 12
Response (kDAC)
Bias (DAC)
20
0
−20
−40
−60
0
250
500
750
1000
1250
1500
1750
2000
c12r04
c12r07
c12r08
c12r15
c12r18
c12r20
Best Bias (155)
User Bias (150)

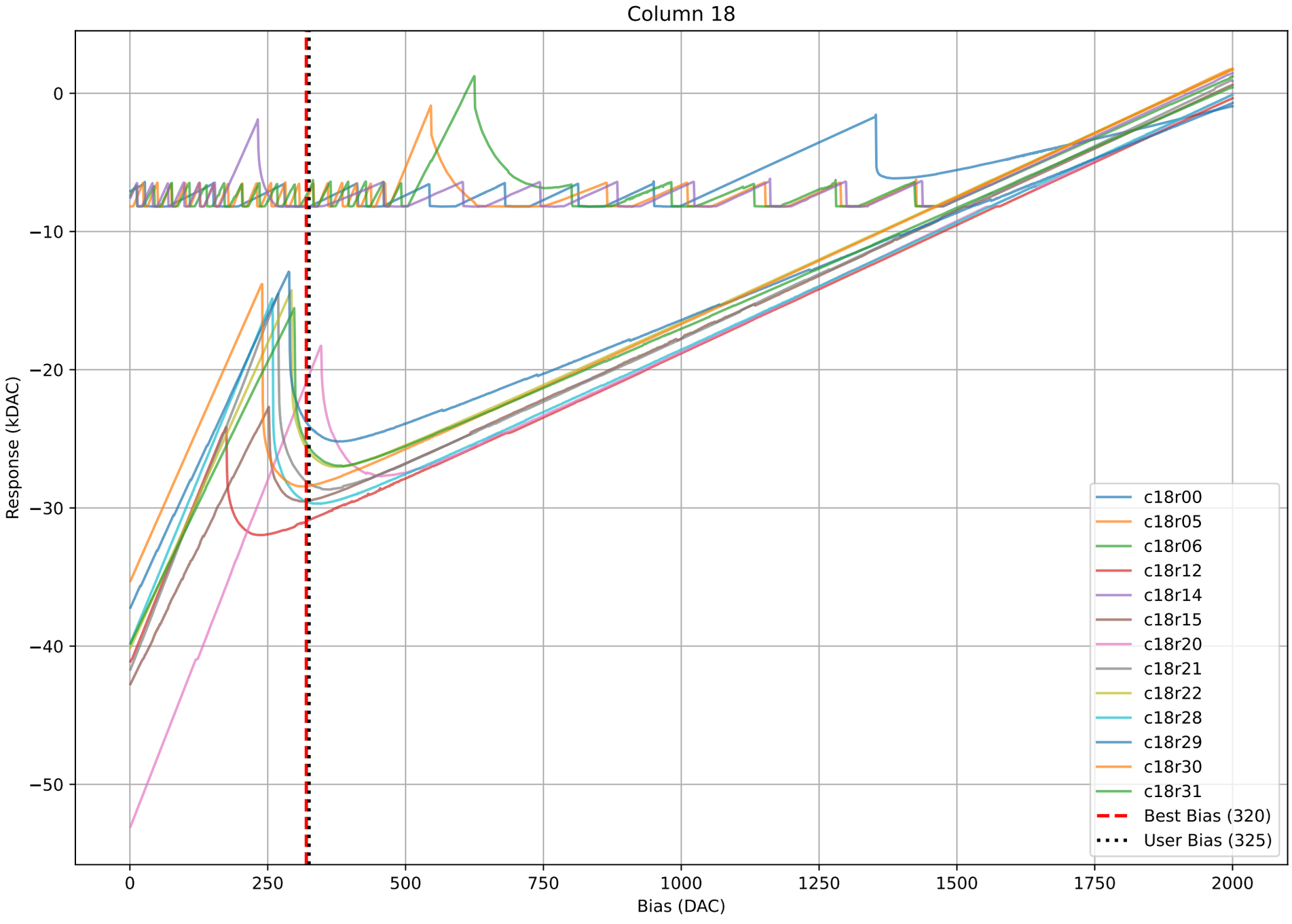

Column 13
Response (kDAC)
4.5
4.0
3.5
3.0
2.5
2.0
1.5
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
c13r21
Best Bias (81)
User Bias (0)

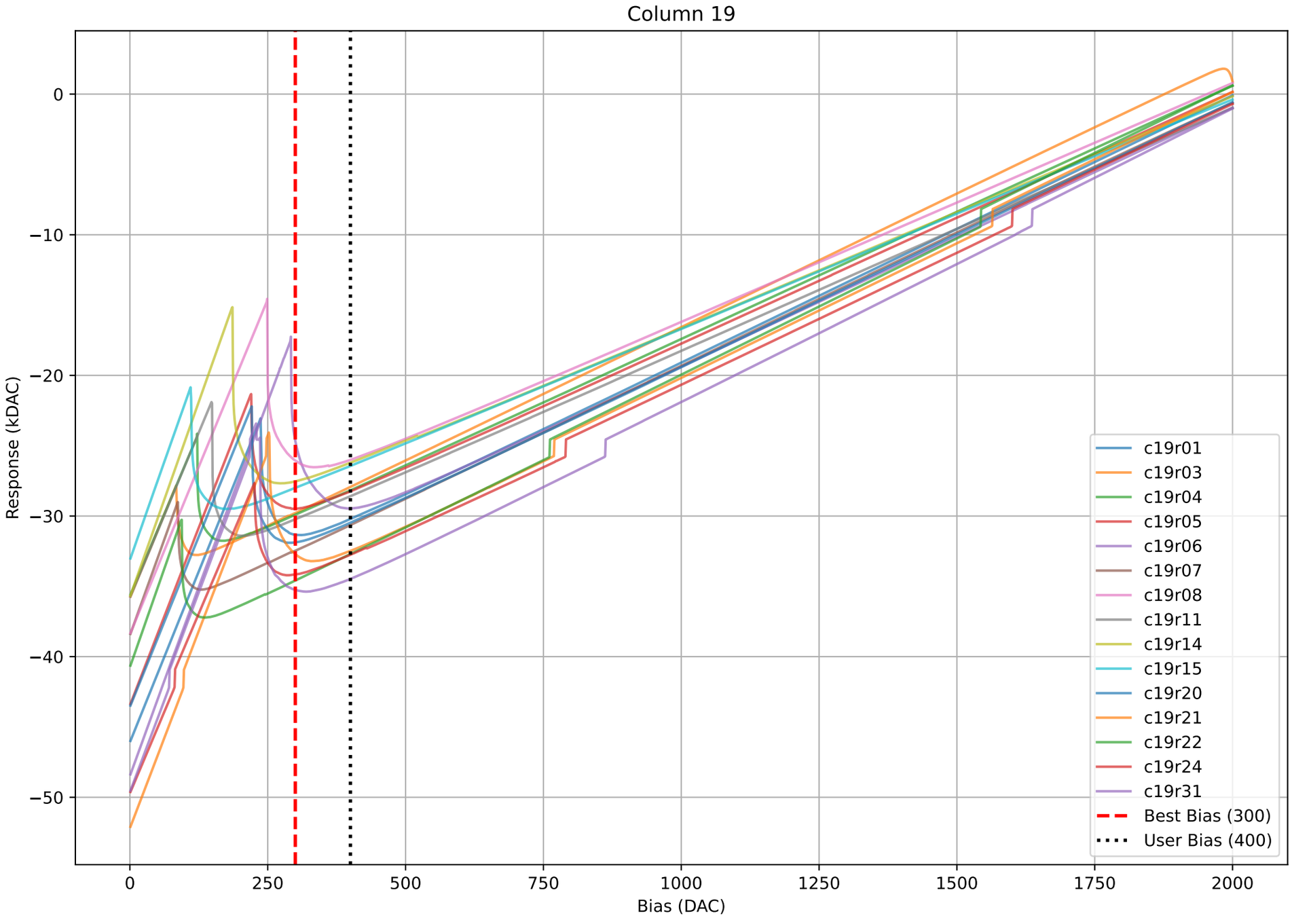

Column 14
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

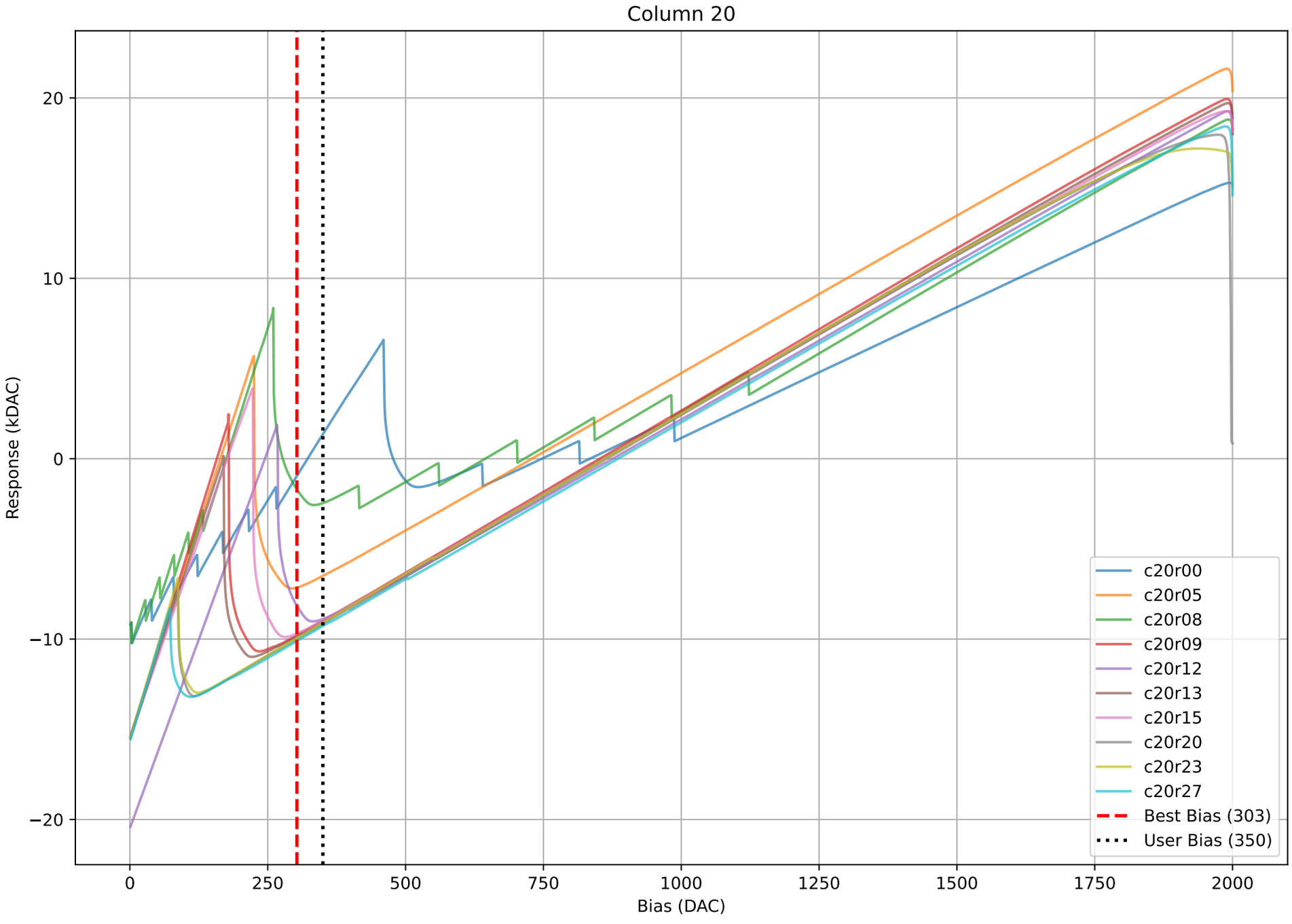

Column 15
Response (kDAC)
Bias (DAC)
20
0
−20
−40
−60
0
250
500
750
1000
1250
1500
1750
2000
c15r01
c15r06
c15r07
c15r08
c15r09
c15r10
c15r12
c15r15
c15r17
c15r20
c15r24
Best Bias (516)
User Bias (800)

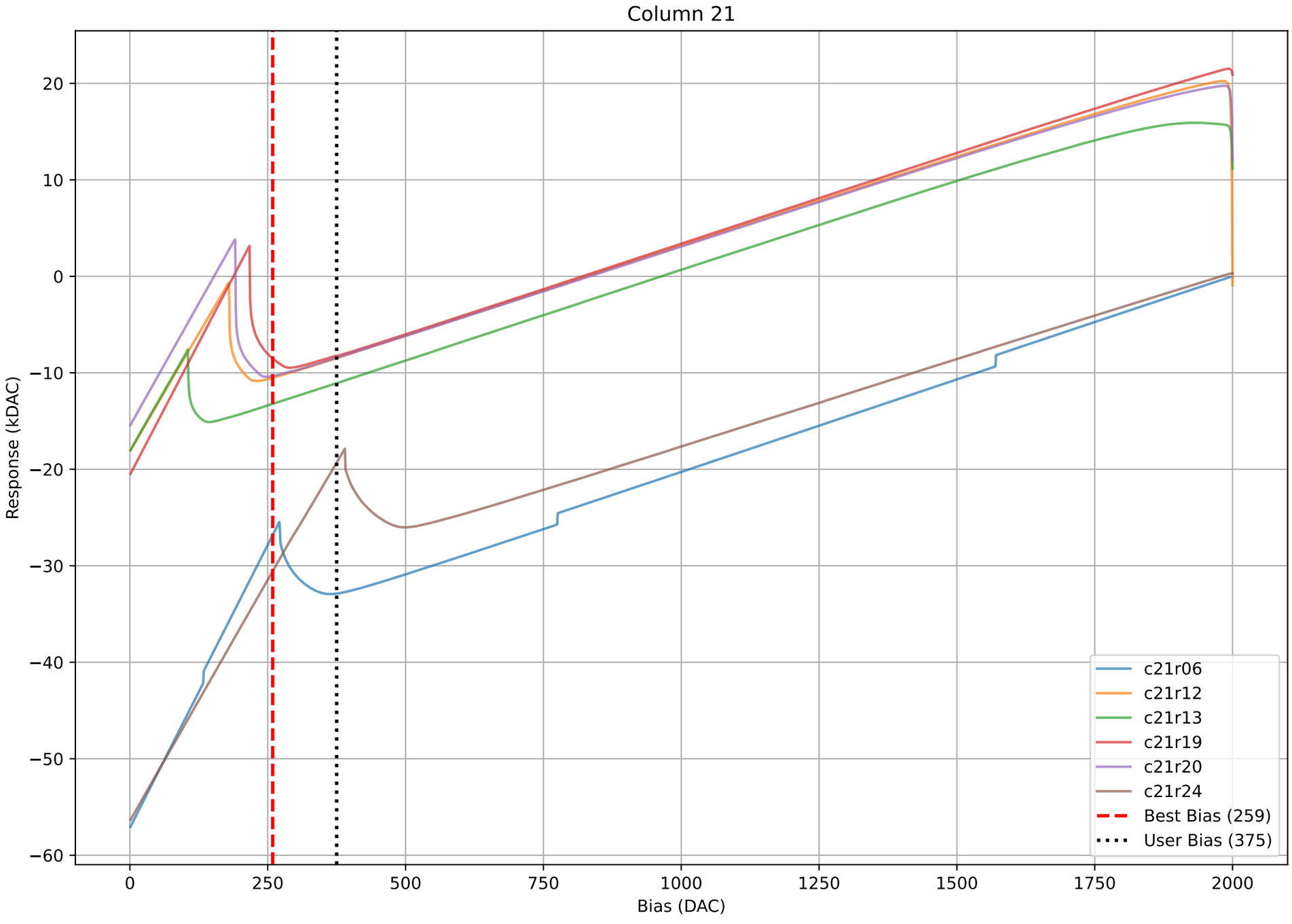

Column 16
c16r03
c16r12
c16r30
Best Bias (967)
User Bias (550)
Response (kDAC)
Bias (DAC)
10
5
0
−5
−10
−15
0
250
500
750
1000
1250
1500
1750
2000

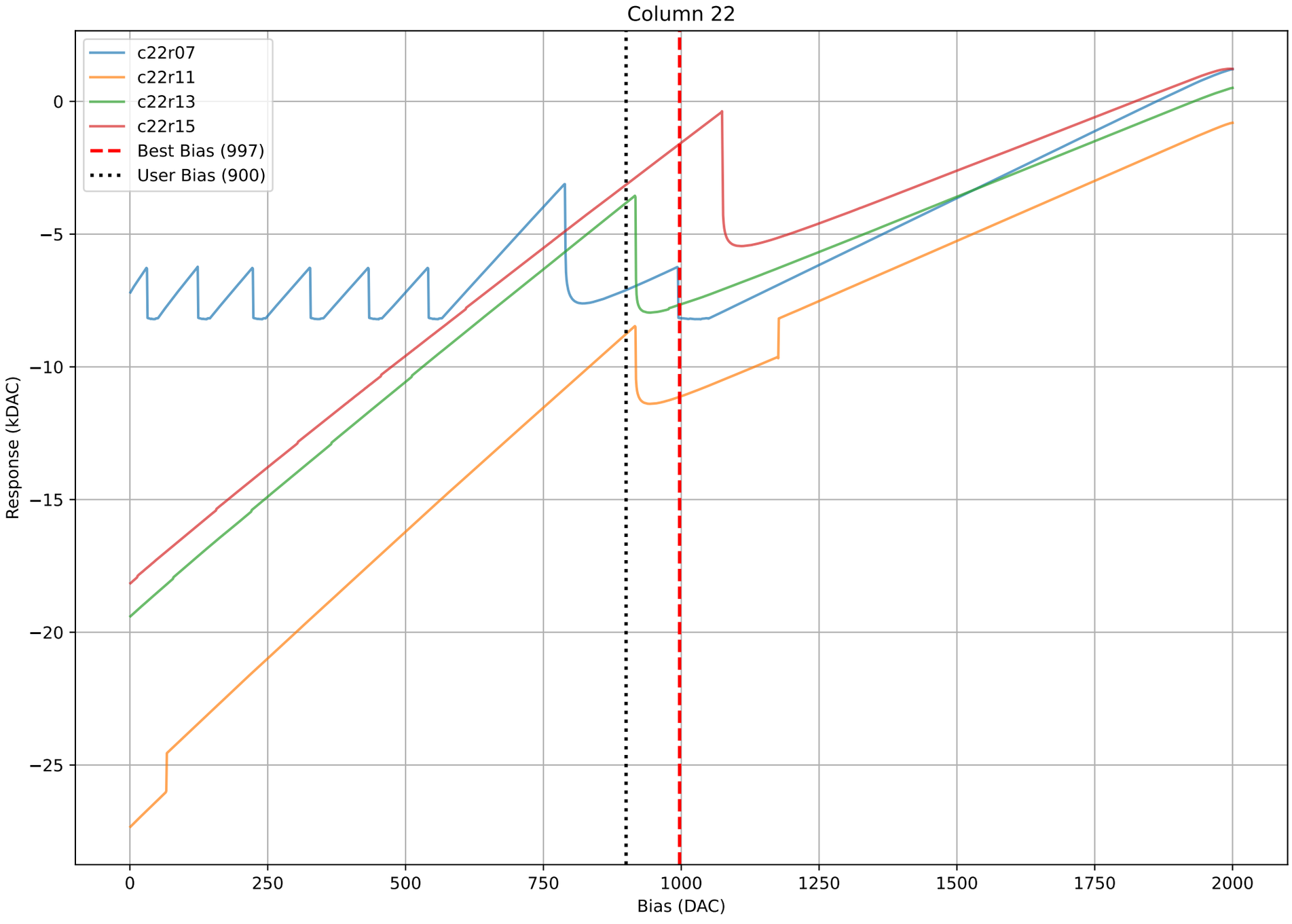

Column 17
Response (kDAC)
Bias (DAC)
5
0
−5
−10
−15
−20
−25
−30
0
250
500
750
1000
1250
1500
1750
2000
c17r08
c17r32
Best Bias (876)
User Bias (0)

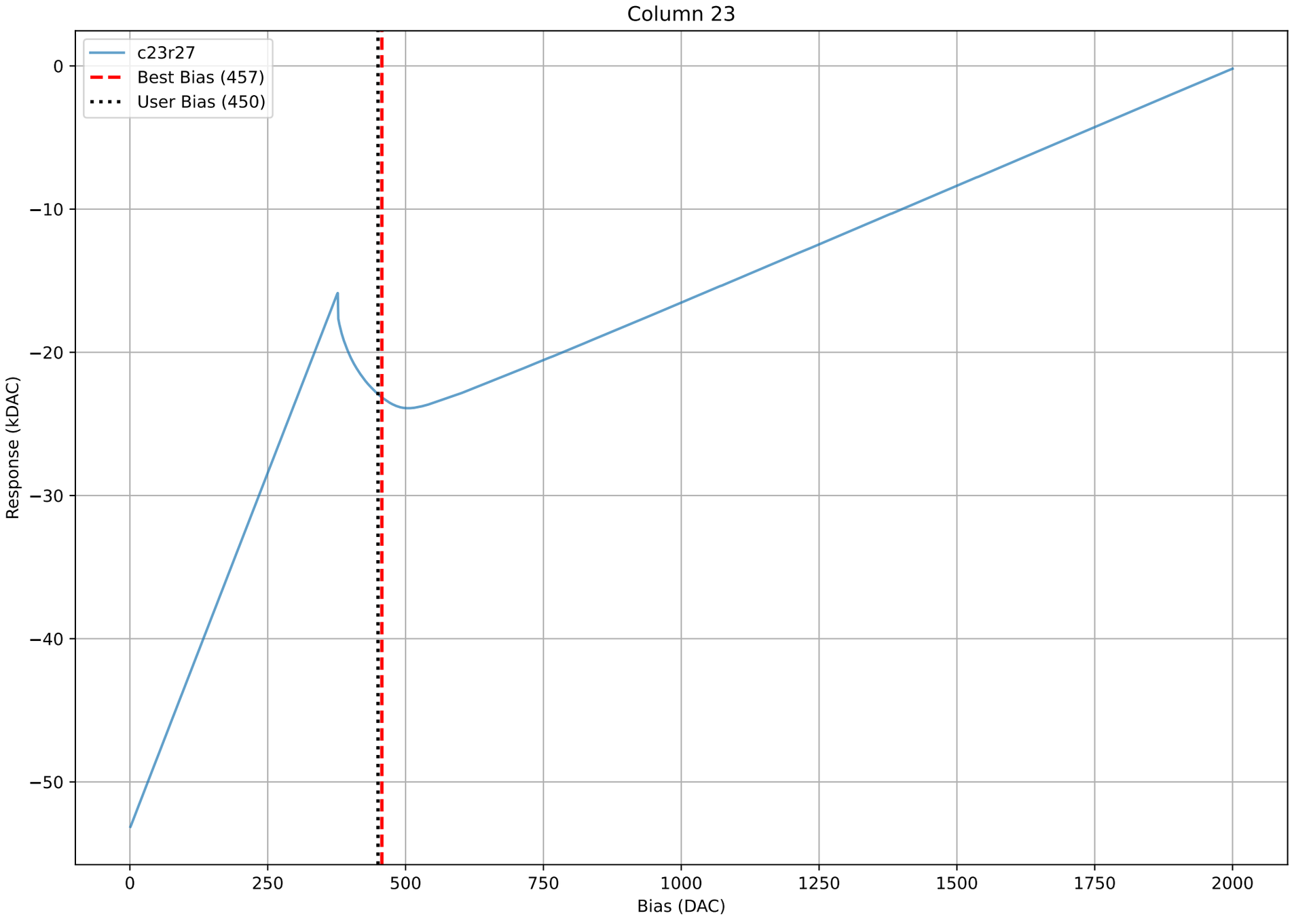

Column 18
Response (kDAC)
Bias (DAC)
0
−10
−20
−30
−40
−50
0
250
500
750
1000
1250
1500
1750
2000
c18r00
c18r05
c18r06
c18r12
c18r14
c18r15
c18r20
c18r21
c18r22
c18r28
c18r29
c18r30
c18r31
Best Bias (320)
User Bias (325)

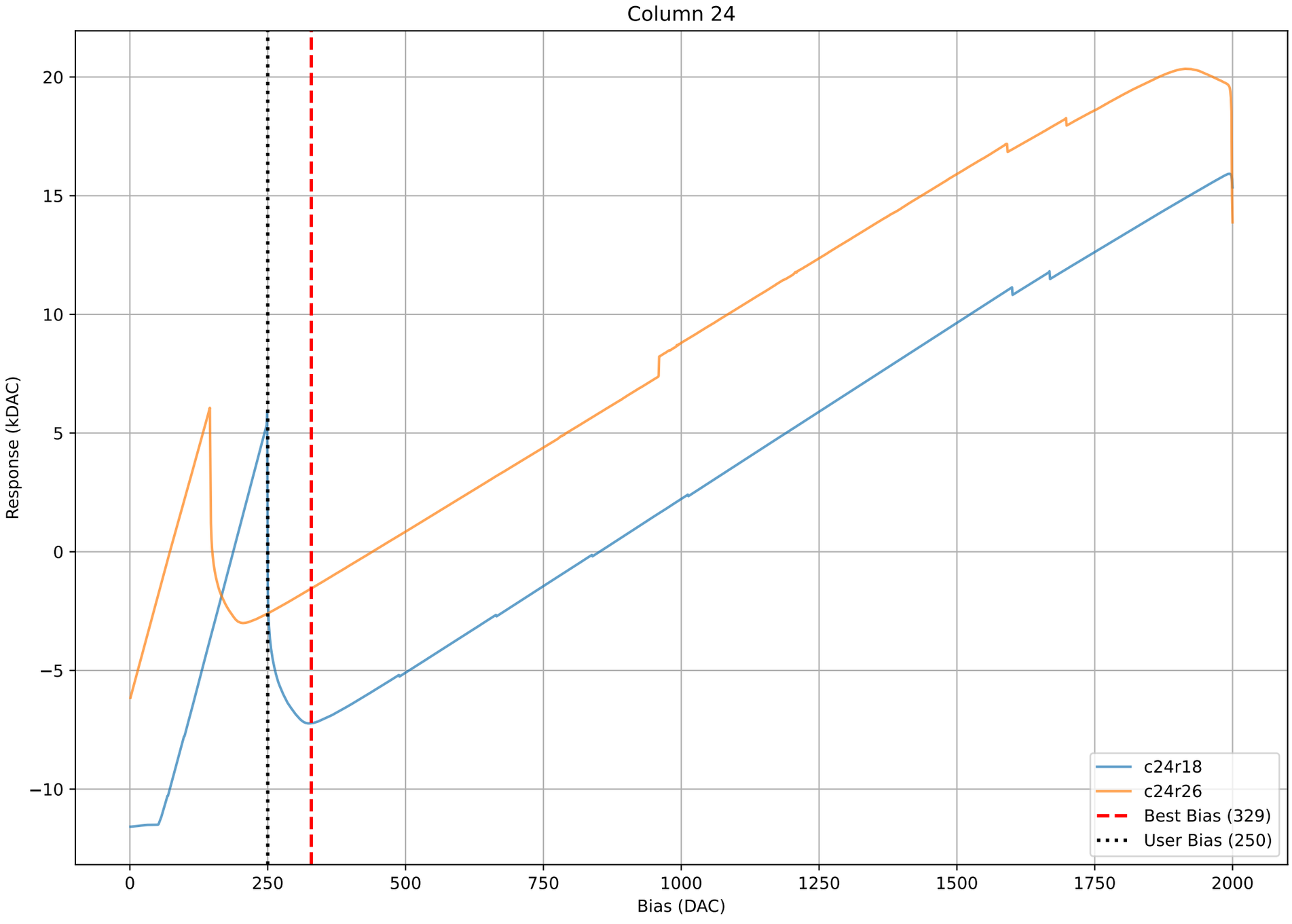

Column 19
Response (kDAC)
Bias (DAC)
0
−10
−20
−30
−40
−50
0
250
500
750
1000
1250
1500
1750
2000
c19r01
c19r03
c19r04
c19r05
c19r06
c19r07
c19r08
c19r11
c19r14
c19r15
c19r20
c19r21
c19r22
c19r24
c19r31
Best Bias (300)
User Bias (400)

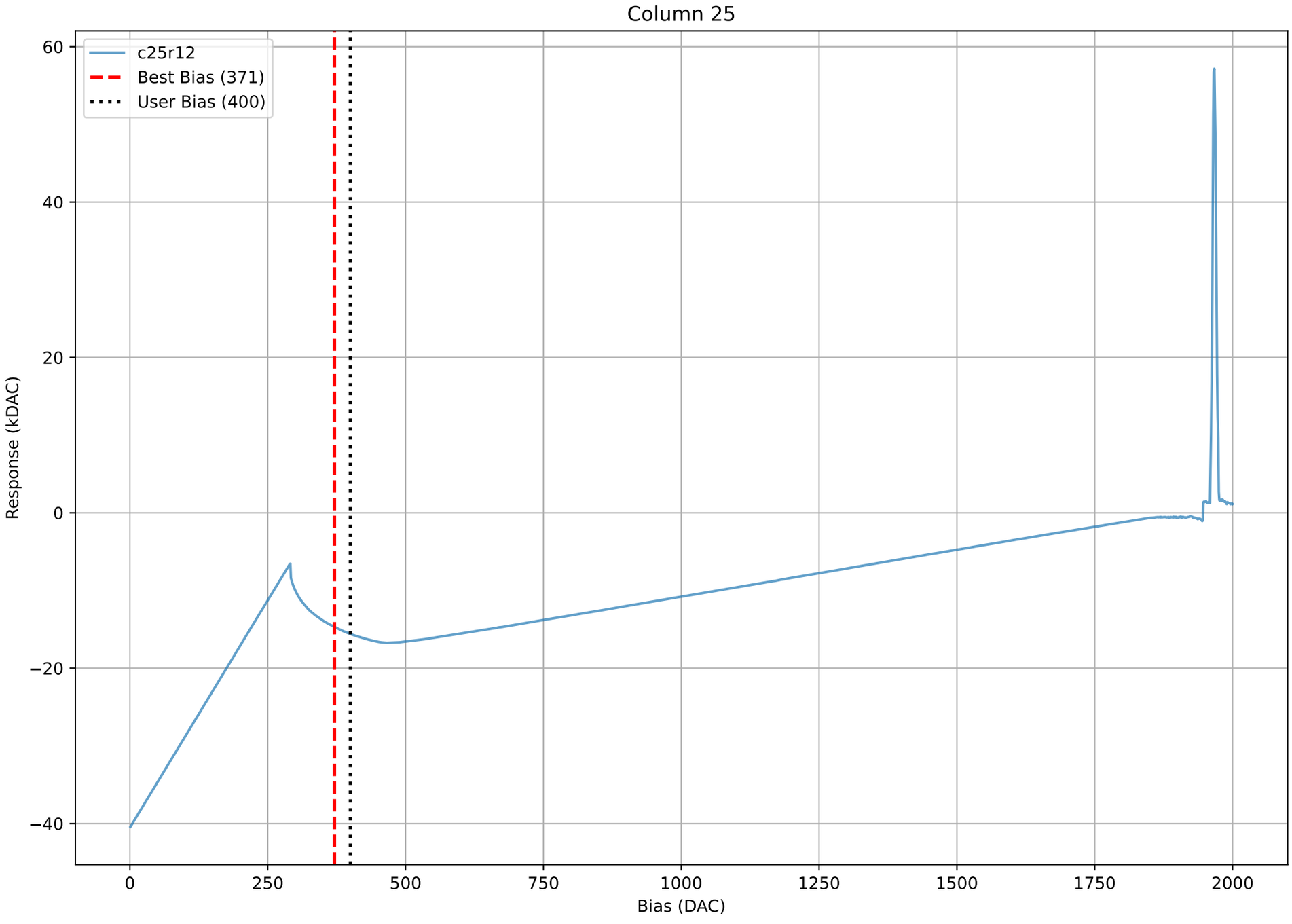

Column 20
Response (kDAC)
Bias (DAC)
20
10
0
−10
−20
0
250
500
750
1000
1250
1500
1750
2000
c20r00
c20r05
c20r08
c20r09
c20r12
c20r13
c20r15
c20r20
c20r23
c20r27
Best Bias (303)
User Bias (350)

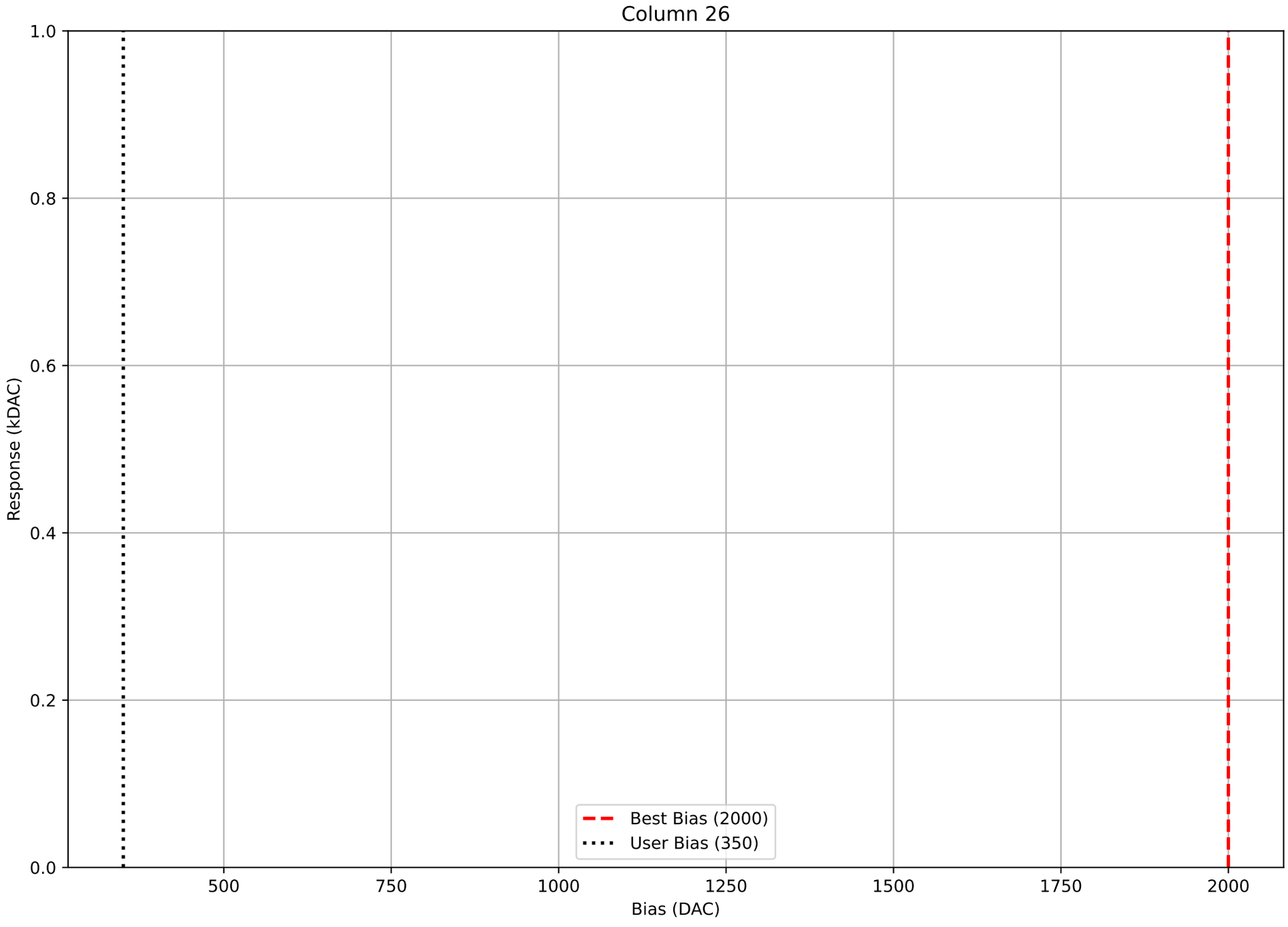

Column 21
Response (kDAC)
Bias (DAC)
20
10
0
−10
−20
−30
−40
−50
−60
0
250
500
750
1000
1250
1500
1750
2000
c21r06
c21r12
c21r13
c21r19
c21r20
c21r24
Best Bias (259)
User Bias (375)

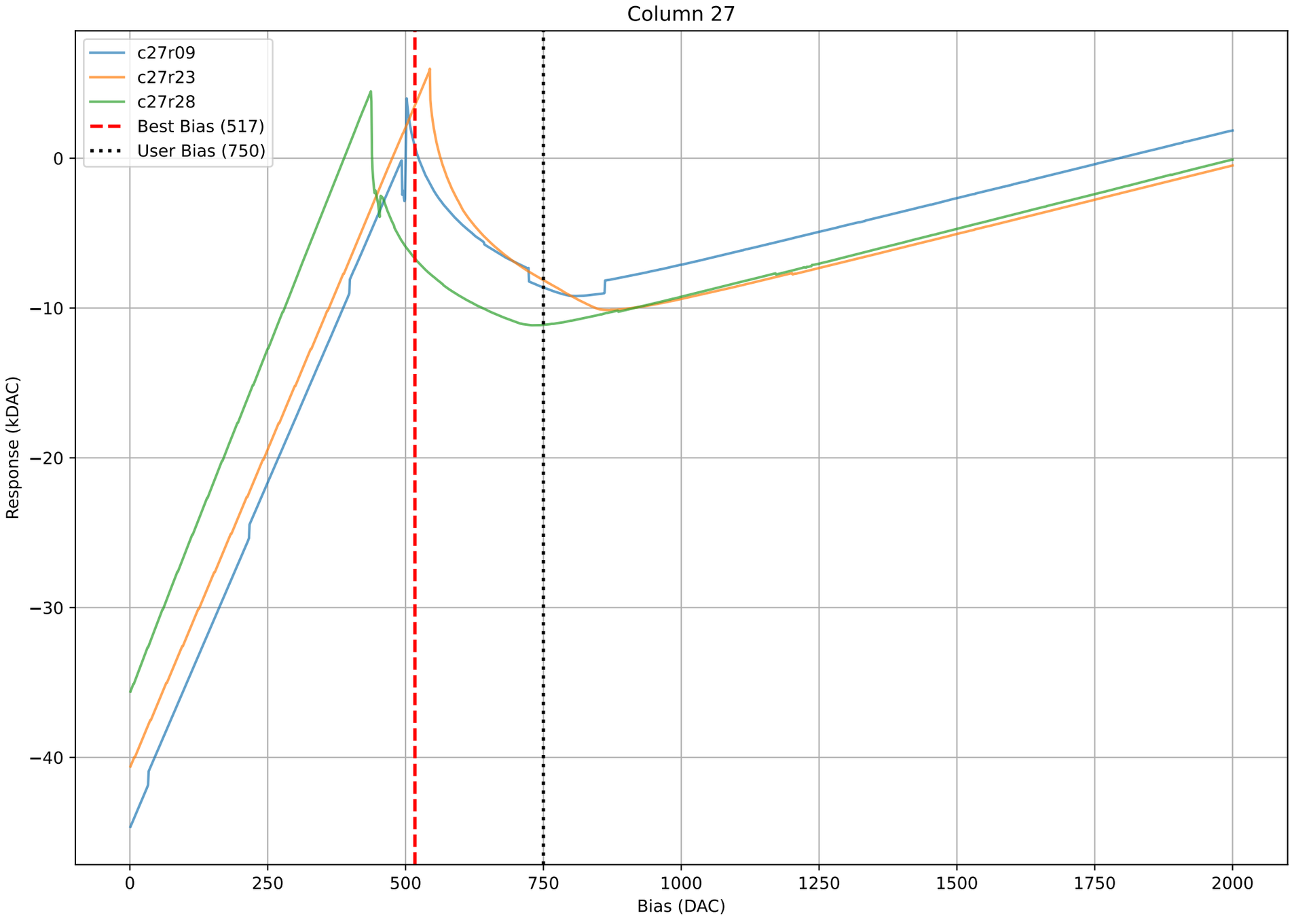

Column 22
c22r07
c22r11
c22r13
c22r15
Best Bias (997)
User Bias (900)
Response (kDAC)
0
−5
−10
−15
−20
−25
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)

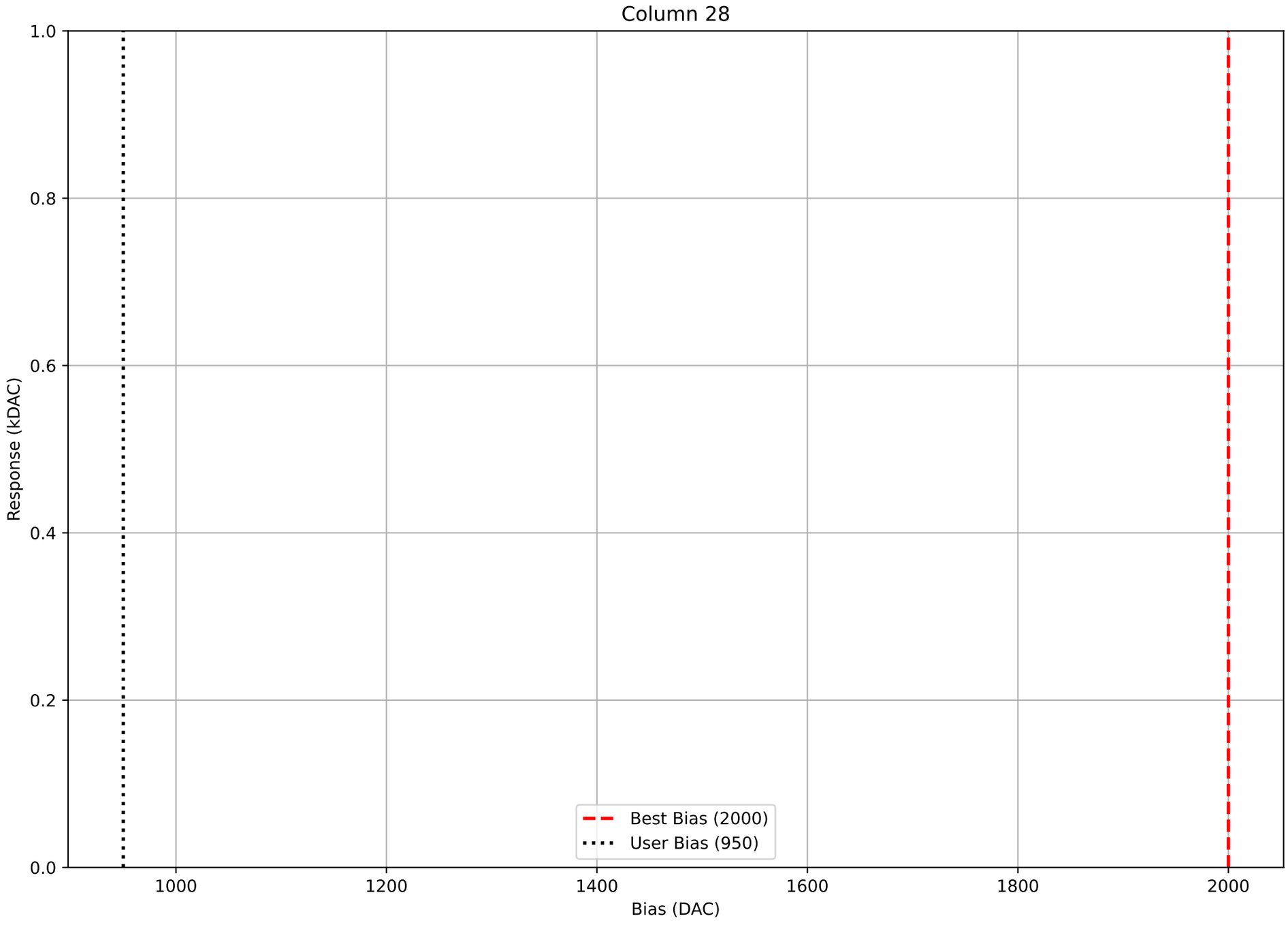

Column 23
c23r27
Best Bias (457)
User Bias (450)
0
−10
−20
−30
−40
−50
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)

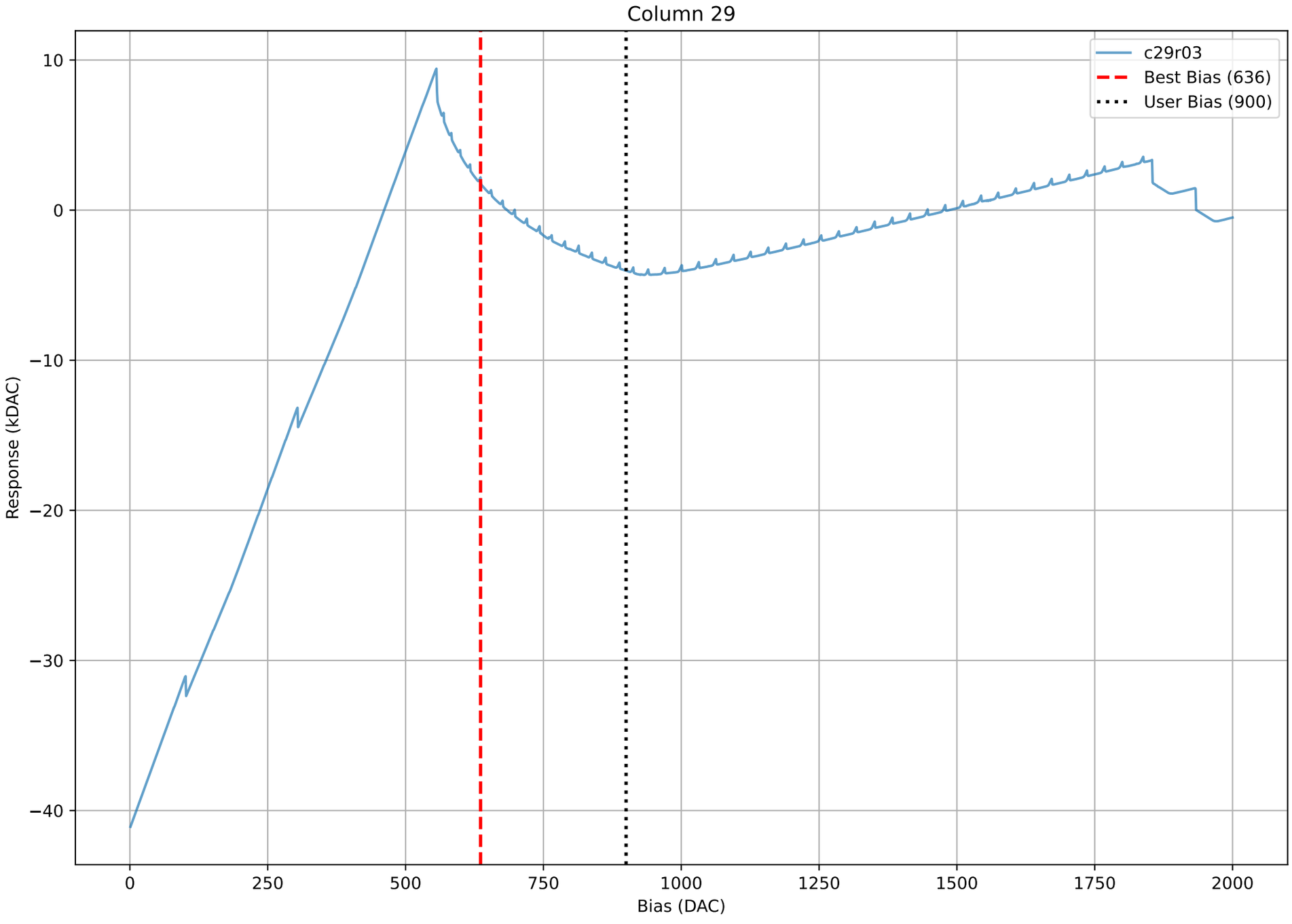

Column 24
Response (kDAC)
Bias (DAC)
20
15
10
5
0
−5
−10
0
250
500
750
1000
1250
1500
1750
2000
c24r18
c24r26
Best Bias (329)
User Bias (250)

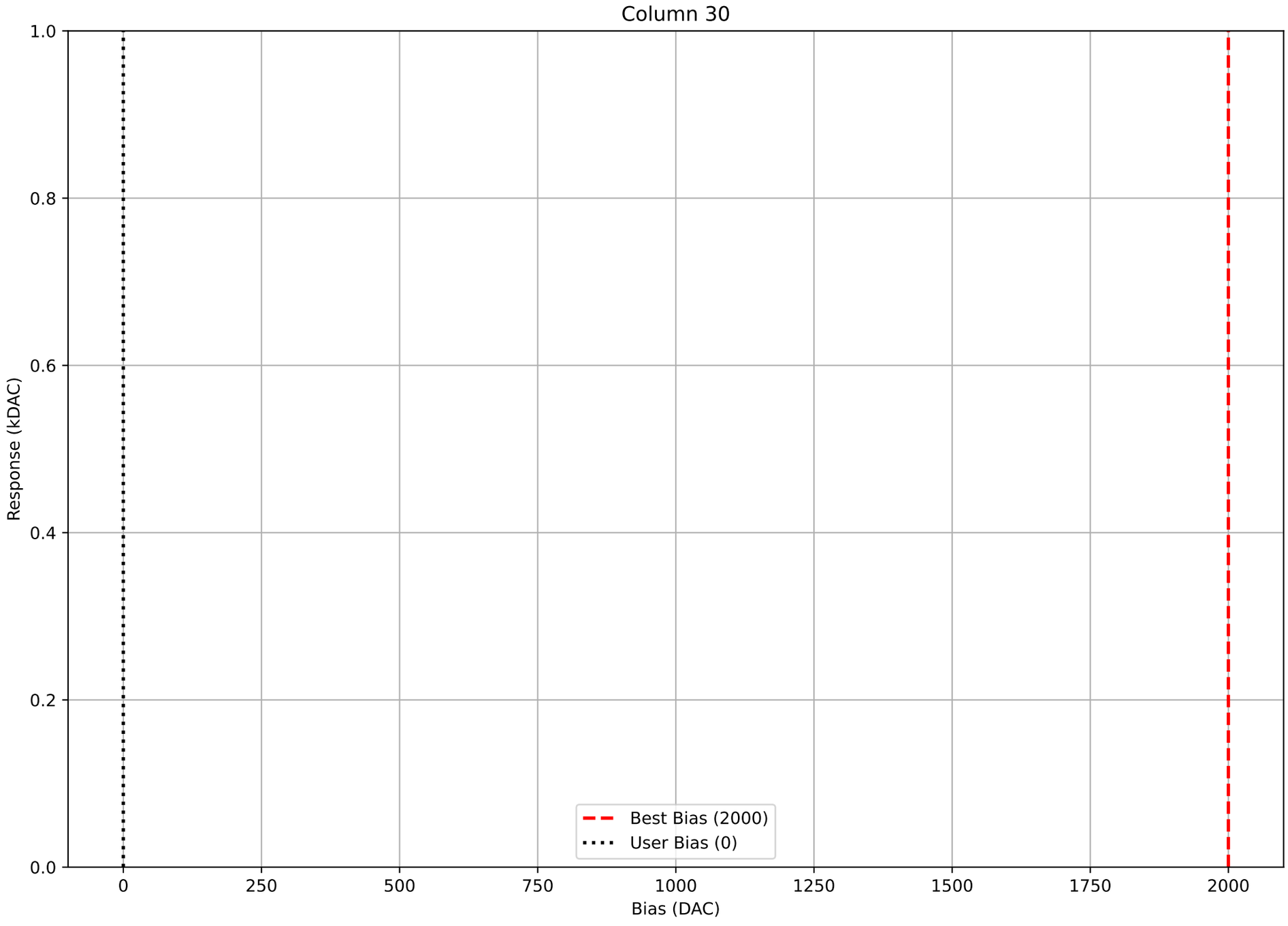

Column 25
c25r12
Best Bias (371)
User Bias (400)
Response (kDAC)
60
40
20
0
−20
−40
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)

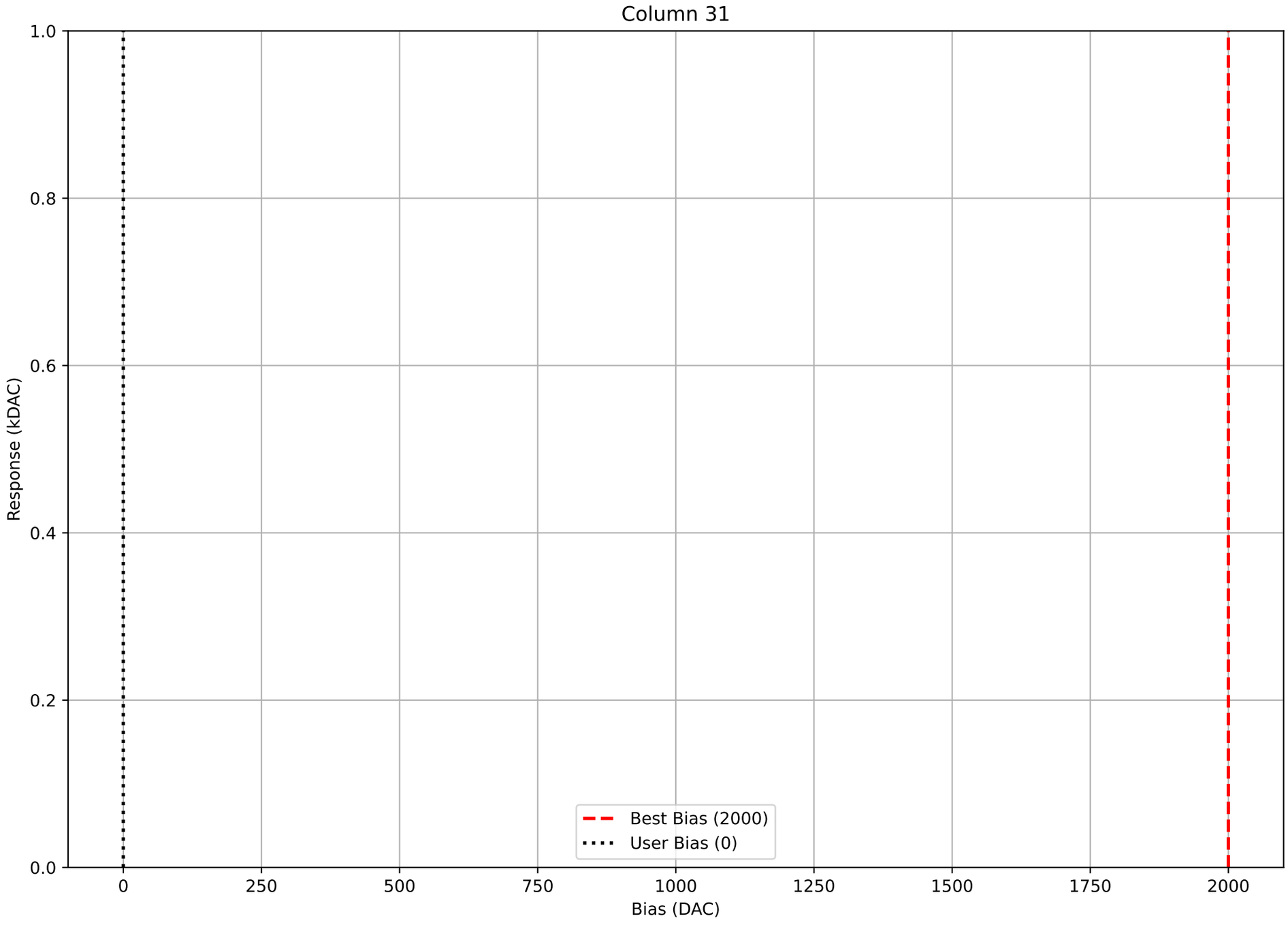

Column 26
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (350)


Column 27
c27r09
c27r23
c27r28
Best Bias (517)
User Bias (750)
0
−10
−20
−30
−40
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)


Column 28
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
1000
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (950)


Column 29
c29r03
Best Bias (636)
User Bias (900)
Response (kDAC)
Bias (DAC)
10
0
−10
−20
−30
−40
0
250
500
750
1000
1250
1500
1750
2000


Column 30
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)


Column 31
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

```
Column 0
  Noisy detectors excluded: []
  Best bias = 514.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [434.0, 514.0]: [(1, 0), (16, 0)]
    row=1 col=0 transition=[453.0, 434.0] min_slope=-3.372 mean_slope=-0.574 drop_kdac=-11.454
    row=16 col=0 transition=[444.0, 434.0] min_slope=-1.778 mean_slope=-0.653 drop_kdac=-7.136
  User bias = 425
  Clean detectors in transition in [345.0, 425.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=1 col=0 transition=[453.0, 434.0] min_slope=-3.372 mean_slope=-0.574 drop_kdac=-11.454
gap_to_window=28.0 reason=accepted transition outside user window
    row=16 col=0 transition=[444.0, 434.0] min_slope=-1.778 mean_slope=-0.653 drop_kdac=-7.136
gap_to_window=19.0 reason=accepted transition outside user window

Column 1
  Noisy detectors excluded: []
  Best bias = 422.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [342.0, 422.0]: [(21, 1), (22, 1)]
    row=21 col=1 transition=[366.0, 342.0] min_slope=-3.743 mean_slope=-0.531 drop_kdac=-13.213
    row=22 col=1 transition=[398.0, 378.0] min_slope=-4.581 mean_slope=-0.711 drop_kdac=-14.934
  User bias = 400
  Clean detectors in transition in [320.0, 400.0]: [(21, 1), (22, 1)]
    row=21 col=1 transition=[366.0, 342.0] min_slope=-3.743 mean_slope=-0.531 drop_kdac=-13.213
    row=22 col=1 transition=[398.0, 378.0] min_slope=-4.581 mean_slope=-0.711 drop_kdac=-14.934
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 2
  Noisy detectors excluded: [(14, 2)]
    row=14 col=2 noisy_transition=[1330.0, 1326.0] min_slope=-1.299 mean_slope=-0.545 drop_kdac=-2.721
    row=14 col=2 noisy_transition=[1312.0, 1310.0] min_slope=-1.288 mean_slope=-0.869 drop_kdac=-2.575
    row=14 col=2 noisy_transition=[850.0, 846.0] min_slope=-12.697 mean_slope=-9.052 drop_kdac=-38.998
    row=14 col=2 noisy_transition=[843.0, 838.0] min_slope=-11.190 mean_slope=-6.686 drop_kdac=-35.508
    row=14 col=2 noisy_transition=[809.0, 806.0] min_slope=-5.587 mean_slope=-3.479 drop_kdac=-13.997
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 425
  Clean detectors in transition in [345.0, 425.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=13 col=2 transition=[435.0, 435.0] min_slope=-0.128 mean_slope=-0.128 drop_kdac=0.000
gap_to_window=10.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
```

```
segment too short for sustained rule
    row=16 col=2 transition=[436.0, 435.0] min_slope=-0.139 mean_slope=-0.135 drop_kdac=-0.281
gap_to_window=11.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 3
  Noisy detectors excluded: []
  Best bias = 364.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [284.0, 364.0]: [(1, 3), (11, 3)]
    row=1 col=3 transition=[303.0, 284.0] min_slope=-3.873 mean_slope=-0.611 drop_kdac=-12.318
    row=11 col=3 transition=[351.0, 327.0] min_slope=-4.311 mean_slope=-0.636 drop_kdac=-15.919
  User bias = 350
  Clean detectors in transition in [270.0, 350.0]: [(1, 3)]
    row=1 col=3 transition=[303.0, 284.0] min_slope=-3.873 mean_slope=-0.611 drop_kdac=-12.318
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=11 col=3 transition=[351.0, 327.0] min_slope=-4.311 mean_slope=-0.636 drop_kdac=-15.919
gap_to_window=1.0 reason=accepted transition outside user window
    row=12 col=3 transition=[254.0, 238.0] min_slope=-3.663 mean_slope=-0.672 drop_kdac=-11.437
gap_to_window=32.0 reason=accepted transition outside user window
    row=16 col=3 transition=[259.0, 258.0] min_slope=-0.154 mean_slope=-0.154 drop_kdac=-0.333
gap_to_window=12.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 4
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 5
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 6
  Noisy detectors excluded: []
  Best bias = 2000.0
```

```
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 7
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 8
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 9
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 10
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
```

```
    []

Column 11
  Noisy detectors excluded: [(3, 11), (6, 11)]
    row=3 col=11 noisy_transition=[1676.0, 1667.0] min_slope=-6.407 mean_slope=-3.651 drop_kdac=-36.416
    row=3 col=11 noisy_transition=[258.0, 257.0] min_slope=-1.012 mean_slope=-0.798 drop_kdac=-1.346
    row=6 col=11 noisy_transition=[1315.0, 1314.0] min_slope=-1.496 mean_slope=-1.474 drop_kdac=-2.940
    row=6 col=11 noisy_transition=[681.0, 677.0] min_slope=-1.562 mean_slope=-1.191 drop_kdac=-5.979
    row=6 col=11 noisy_transition=[583.0, 578.0] min_slope=-1.502 mean_slope=-0.475 drop_kdac=-2.675
    row=6 col=11 noisy_transition=[569.0, 567.0] min_slope=-1.513 mean_slope=-0.978 drop_kdac=-3.027
    row=6 col=11 noisy_transition=[176.0, 140.0] min_slope=-4.193 mean_slope=-0.530 drop_kdac=-19.726
    row=6 col=11 noisy_transition=[95.0, 94.0] min_slope=-1.331 mean_slope=-1.284 drop_kdac=-2.681
    row=6 col=11 noisy_transition=[79.0, 78.0] min_slope=-2.913 mean_slope=-2.846 drop_kdac=-5.819
  Best bias = 1183.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [1103.0, 1183.0]: [(16, 11)]
    row=16 col=11 transition=[1105.0, 1103.0] min_slope=-1.018 mean_slope=-0.689 drop_kdac=-2.037
  User bias = 150
  Clean detectors in transition in [70.0, 150.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=6 col=11 transition=[176.0, 140.0] min_slope=-4.193 mean_slope=-0.530 drop_kdac=-19.726
gap_to_window=26.0 reason=excluded as noisy (multiple accepted transitions)
    row=6 col=11 transition=[95.0, 94.0] min_slope=-1.331 mean_slope=-1.284 drop_kdac=-2.681 gap_to_window=0.0
reason=excluded as noisy (multiple accepted transitions)
    row=6 col=11 transition=[79.0, 78.0] min_slope=-2.913 mean_slope=-2.846 drop_kdac=-5.819 gap_to_window=0.0
reason=excluded as noisy (multiple accepted transitions)

Column 12
  Noisy detectors excluded: []
  Best bias = 155.0
  Max detectors in transition at best bias = 5
  Detectors at best bias in [75.0, 155.0]: [(4, 12), (7, 12), (8, 12), (15, 12), (20, 12)]
    row=4 col=12 transition=[125.0, 105.0] min_slope=-1.120 mean_slope=-0.306 drop_kdac=-6.558
    row=7 col=12 transition=[94.0, 75.0] min_slope=-1.442 mean_slope=-0.383 drop_kdac=-7.731
    row=8 col=12 transition=[129.0, 109.0] min_slope=-3.119 mean_slope=-0.579 drop_kdac=-12.190
    row=15 col=12 transition=[116.0, 97.0] min_slope=-2.903 mean_slope=-0.512 drop_kdac=-10.297
    row=20 col=12 transition=[138.0, 115.0] min_slope=-3.258 mean_slope=-0.543 drop_kdac=-12.964
  User bias = 150
  Clean detectors in transition in [70.0, 150.0]: [(4, 12), (7, 12), (8, 12), (15, 12), (20, 12)]
    row=4 col=12 transition=[125.0, 105.0] min_slope=-1.120 mean_slope=-0.306 drop_kdac=-6.558
    row=7 col=12 transition=[94.0, 75.0] min_slope=-1.442 mean_slope=-0.383 drop_kdac=-7.731
    row=8 col=12 transition=[129.0, 109.0] min_slope=-3.119 mean_slope=-0.579 drop_kdac=-12.190
    row=15 col=12 transition=[116.0, 97.0] min_slope=-2.903 mean_slope=-0.512 drop_kdac=-10.297
    row=20 col=12 transition=[138.0, 115.0] min_slope=-3.258 mean_slope=-0.543 drop_kdac=-12.964
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
```

```
    row=13 col=12 transition=[75.0, 60.0] min_slope=-0.845 mean_slope=-0.282 drop_kdac=-4.503
gap_to_window=10.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=24 col=12 transition=[66.0, 51.0] min_slope=-0.727 mean_slope=-0.273 drop_kdac=-4.299
gap_to_window=19.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=28 col=12 transition=[45.0, 35.0] min_slope=-0.219 mean_slope=-0.144 drop_kdac=-1.495
gap_to_window=35.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 13
  Noisy detectors excluded: [(3, 13), (4, 13), (5, 13), (6, 13)]
    row=3 col=13 noisy_transition=[1361.0, 1359.0] min_slope=-2.260 mean_slope=-1.506 drop_kdac=-4.520
    row=3 col=13 noisy_transition=[905.0, 903.0] min_slope=-2.270 mean_slope=-1.517 drop_kdac=-4.541
    row=3 col=13 noisy_transition=[590.0, 588.0] min_slope=-2.256 mean_slope=-1.500 drop_kdac=-4.512
    row=3 col=13 noisy_transition=[288.0, 286.0] min_slope=-2.287 mean_slope=-1.531 drop_kdac=-4.573
    row=4 col=13 noisy_transition=[911.0, 909.0] min_slope=-2.145 mean_slope=-1.341 drop_kdac=-4.290
    row=4 col=13 noisy_transition=[592.0, 588.0] min_slope=-2.243 mean_slope=-1.020 drop_kdac=-5.072
    row=4 col=13 noisy_transition=[279.0, 277.0] min_slope=-2.204 mean_slope=-1.470 drop_kdac=-4.409
    row=4 col=13 noisy_transition=[103.0, 101.0] min_slope=-1.396 mean_slope=-0.651 drop_kdac=-2.792
    row=4 col=13 noisy_transition=[25.0, 23.0] min_slope=-2.259 mean_slope=-1.496 drop_kdac=-4.518
    row=5 col=13 noisy_transition=[868.0, 866.0] min_slope=-2.096 mean_slope=-1.396 drop_kdac=-4.191
    row=5 col=13 noisy_transition=[547.0, 543.0] min_slope=-2.110 mean_slope=-1.324 drop_kdac=-6.656
    row=5 col=13 noisy_transition=[236.0, 235.0] min_slope=-1.840 mean_slope=-1.778 drop_kdac=-3.378
    row=6 col=13 noisy_transition=[1192.0, 1190.0] min_slope=-2.035 mean_slope=-1.358 drop_kdac=-4.070
    row=6 col=13 noisy_transition=[796.0, 794.0] min_slope=-1.991 mean_slope=-1.331 drop_kdac=-3.981
    row=6 col=13 noisy_transition=[488.0, 486.0] min_slope=-2.056 mean_slope=-1.365 drop_kdac=-4.112
    row=6 col=13 noisy_transition=[189.0, 187.0] min_slope=-2.047 mean_slope=-1.360 drop_kdac=-4.094
  Best bias = 81.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [1.0, 81.0]: [(21, 13)]
    row=21 col=13 transition=[2.0, 1.0] min_slope=-1.059 mean_slope=-0.718 drop_kdac=-1.059
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=4 col=13 transition=[25.0, 23.0] min_slope=-2.259 mean_slope=-1.496 drop_kdac=-4.518
gap_to_window=25.0 reason=excluded as noisy (multiple accepted transitions)
    row=18 col=13 transition=[6.0, 5.0] min_slope=-0.349 mean_slope=-0.337 drop_kdac=-0.652 gap_to_window=6.0
reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=19 col=13 transition=[5.0, 2.0] min_slope=-0.340 mean_slope=-0.235 drop_kdac=-1.009 gap_to_window=5.0
reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=20 col=13 transition=[40.0, 39.0] min_slope=-0.357 mean_slope=-0.329 drop_kdac=-0.652
gap_to_window=40.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=20 col=13 transition=[28.0, 27.0] min_slope=-0.328 mean_slope=-0.327 drop_kdac=-0.684
gap_to_window=28.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=20 col=13 transition=[15.0, 14.0] min_slope=-0.337 mean_slope=-0.328 drop_kdac=-0.632
gap_to_window=15.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=20 col=13 transition=[3.0, 1.0] min_slope=-0.423 mean_slope=-0.332 drop_kdac=-0.846 gap_to_window=3.0
```

```
reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=21 col=13 transition=[2.0, 1.0] min_slope=-1.059 mean_slope=-0.718 drop_kdac=-1.059 gap_to_window=2.0
reason=accepted transition outside user window
    row=22 col=13 transition=[38.0, 38.0] min_slope=-0.083 mean_slope=-0.083 drop_kdac=0.000
gap_to_window=38.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=22 col=13 transition=[26.0, 26.0] min_slope=-0.122 mean_slope=-0.122 drop_kdac=0.000
gap_to_window=26.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=22 col=13 transition=[13.0, 13.0] min_slope=-0.142 mean_slope=-0.142 drop_kdac=0.000
gap_to_window=13.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=22 col=13 transition=[7.0, 7.0] min_slope=-0.260 mean_slope=-0.260 drop_kdac=0.000 gap_to_window=7.0
reason=slope not steep enough for sharp rule, total drop too small for sustained rule, segment too short for
sustained rule
    row=22 col=13 transition=[1.0, 1.0] min_slope=-0.190 mean_slope=-0.190 drop_kdac=0.000 gap_to_window=1.0
reason=slope not steep enough for sharp rule, total drop too small for sustained rule, segment too short for
sustained rule
    row=23 col=13 transition=[38.0, 38.0] min_slope=-0.131 mean_slope=-0.131 drop_kdac=0.000
gap_to_window=38.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=23 col=13 transition=[29.0, 29.0] min_slope=-0.087 mean_slope=-0.087 drop_kdac=0.000
gap_to_window=29.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=23 col=13 transition=[26.0, 26.0] min_slope=-0.090 mean_slope=-0.090 drop_kdac=0.000
gap_to_window=26.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=23 col=13 transition=[16.0, 13.0] min_slope=-0.357 mean_slope=-0.157 drop_kdac=-0.630
gap_to_window=16.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=23 col=13 transition=[3.0, 2.0] min_slope=-0.206 mean_slope=-0.186 drop_kdac=-0.517 gap_to_window=3.0
reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=24 col=13 transition=[26.0, 25.0] min_slope=-0.440 mean_slope=-0.369 drop_kdac=-0.733
gap_to_window=26.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=24 col=13 transition=[14.0, 13.0] min_slope=-0.506 mean_slope=-0.407 drop_kdac=-0.765
gap_to_window=14.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=24 col=13 transition=[2.0, 1.0] min_slope=-0.659 mean_slope=-0.403 drop_kdac=-0.659 gap_to_window=2.0
reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 14
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
```

```
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 15
  Noisy detectors excluded: [(3, 15), (4, 15), (5, 15), (21, 15)]
    row=3 col=15 noisy_transition=[586.0, 583.0] min_slope=-6.587 mean_slope=-2.751 drop_kdac=-8.223
    row=3 col=15 noisy_transition=[570.0, 565.0] min_slope=-7.075 mean_slope=-2.471 drop_kdac=-14.856
    row=3 col=15 noisy_transition=[555.0, 552.0] min_slope=-8.151 mean_slope=-3.570 drop_kdac=-16.444
    row=3 col=15 noisy_transition=[545.0, 541.0] min_slope=-2.927 mean_slope=-0.987 drop_kdac=-5.946
    row=3 col=15 noisy_transition=[534.0, 533.0] min_slope=-2.150 mean_slope=-1.250 drop_kdac=-0.926
    row=3 col=15 noisy_transition=[526.0, 524.0] min_slope=-2.323 mean_slope=-0.875 drop_kdac=-4.647
    row=3 col=15 noisy_transition=[521.0, 501.0] min_slope=-6.490 mean_slope=-2.244 drop_kdac=-49.279
    row=3 col=15 noisy_transition=[482.0, 480.0] min_slope=-2.267 mean_slope=-1.300 drop_kdac=-4.533
    row=4 col=15 noisy_transition=[600.0, 598.0] min_slope=-3.459 mean_slope=-2.271 drop_kdac=-6.917
    row=4 col=15 noisy_transition=[582.0, 576.0] min_slope=-5.810 mean_slope=-3.185 drop_kdac=-22.406
    row=4 col=15 noisy_transition=[546.0, 538.0] min_slope=-4.891 mean_slope=-1.162 drop_kdac=-13.068
    row=4 col=15 noisy_transition=[528.0, 527.0] min_slope=-1.181 mean_slope=-0.728 drop_kdac=-0.250
    row=4 col=15 noisy_transition=[520.0, 518.0] min_slope=-2.455 mean_slope=-0.998 drop_kdac=-0.797
    row=4 col=15 noisy_transition=[514.0, 510.0] min_slope=-2.232 mean_slope=-0.851 drop_kdac=-4.338
    row=4 col=15 noisy_transition=[503.0, 502.0] min_slope=-2.656 mean_slope=-1.864 drop_kdac=-3.472
    row=5 col=15 noisy_transition=[557.0, 547.0] min_slope=-6.675 mean_slope=-2.468 drop_kdac=-27.219
    row=5 col=15 noisy_transition=[471.0, 465.0] min_slope=-1.244 mean_slope=-0.531 drop_kdac=-3.643
    row=21 col=15 noisy_transition=[906.0, 904.0] min_slope=-1.382 mean_slope=-0.899 drop_kdac=-2.765
    row=21 col=15 noisy_transition=[884.0, 882.0] min_slope=-2.681 mean_slope=-1.713 drop_kdac=-5.363
    row=21 col=15 noisy_transition=[731.0, 672.0] min_slope=-6.276 mean_slope=-0.697 drop_kdac=-41.825
    row=21 col=15 noisy_transition=[477.0, 474.0] min_slope=-3.756 mean_slope=-2.387 drop_kdac=-9.488
    row=21 col=15 noisy_transition=[448.0, 446.0] min_slope=-1.063 mean_slope=-0.691 drop_kdac=-2.127
  Best bias = 516.0
  Max detectors in transition at best bias = 6
  Detectors at best bias in [436.0, 516.0]: [(1, 15), (6, 15), (7, 15), (8, 15), (10, 15), (20, 15)]
    row=1 col=15 transition=[511.0, 464.0] min_slope=-7.406 mean_slope=-0.510 drop_kdac=-24.773
    row=6 col=15 transition=[467.0, 446.0] min_slope=-6.901 mean_slope=-1.379 drop_kdac=-30.331
    row=7 col=15 transition=[479.0, 472.0] min_slope=-1.757 mean_slope=-0.882 drop_kdac=-6.766
    row=8 col=15 transition=[481.0, 450.0] min_slope=-1.626 mean_slope=-0.256 drop_kdac=-8.244
    row=10 col=15 transition=[442.0, 436.0] min_slope=-1.181 mean_slope=-0.693 drop_kdac=-4.912
    row=20 col=15 transition=[474.0, 440.0] min_slope=-6.174 mean_slope=-0.570 drop_kdac=-19.917
  User bias = 800
  Clean detectors in transition in [720.0, 800.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 16
  Noisy detectors excluded: [(13, 16), (18, 16), (20, 16)]
    row=13 col=16 noisy_transition=[1103.0, 1102.0] min_slope=-1.010 mean_slope=-0.990 drop_kdac=-2.020
    row=13 col=16 noisy_transition=[487.0, 486.0] min_slope=-1.122 mean_slope=-1.089 drop_kdac=-2.247
```

```
    row=13 col=16 noisy_transition=[194.0, 177.0] min_slope=-5.136 mean_slope=-0.720 drop_kdac=-13.118
    row=13 col=16 noisy_transition=[122.0, 121.0] min_slope=-1.090 mean_slope=-1.040 drop_kdac=-2.181
    row=13 col=16 noisy_transition=[95.0, 94.0] min_slope=-1.051 mean_slope=-1.021 drop_kdac=-2.074
    row=13 col=16 noisy_transition=[69.0, 68.0] min_slope=-1.142 mean_slope=-1.087 drop_kdac=-2.283
    row=13 col=16 noisy_transition=[42.0, 41.0] min_slope=-1.030 mean_slope=-1.022 drop_kdac=-2.059
    row=13 col=16 noisy_transition=[15.0, 14.0] min_slope=-1.014 mean_slope=-1.006 drop_kdac=-2.028
    row=18 col=16 noisy_transition=[952.0, 951.0] min_slope=-1.113 mean_slope=-1.065 drop_kdac=-2.228
    row=18 col=16 noisy_transition=[121.0, 120.0] min_slope=-1.026 mean_slope=-1.017 drop_kdac=-2.051
    row=20 col=16 noisy_transition=[1384.0, 1382.0] min_slope=-1.004 mean_slope=-0.667 drop_kdac=-2.008
    row=20 col=16 noisy_transition=[1196.0, 1195.0] min_slope=-1.102 mean_slope=-1.075 drop_kdac=-2.199
    row=20 col=16 noisy_transition=[787.0, 786.0] min_slope=-1.026 mean_slope=-1.011 drop_kdac=-2.050
    row=20 col=16 noisy_transition=[579.0, 577.0] min_slope=-1.013 mean_slope=-0.673 drop_kdac=-2.026
    row=20 col=16 noisy_transition=[355.0, 354.0] min_slope=-1.092 mean_slope=-1.043 drop_kdac=-2.181
    row=20 col=16 noisy_transition=[145.0, 144.0] min_slope=-1.001 mean_slope=-0.991 drop_kdac=-1.962
  Best bias = 967.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [887.0, 967.0]: [(30, 16)]
    row=30 col=16 transition=[888.0, 887.0] min_slope=-2.131 mean_slope=-2.046 drop_kdac=-4.246
  User bias = 550
  Clean detectors in transition in [470.0, 550.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=6 col=16 transition=[500.0, 499.0] min_slope=-0.706 mean_slope=-0.705 drop_kdac=-1.406
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=13 col=16 transition=[487.0, 486.0] min_slope=-1.122 mean_slope=-1.089 drop_kdac=-2.247
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=19 col=16 transition=[544.0, 542.0] min_slope=-0.182 mean_slope=-0.140 drop_kdac=-0.365
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=19 col=16 transition=[524.0, 523.0] min_slope=-0.849 mean_slope=-0.821 drop_kdac=-1.646
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=20 col=16 transition=[579.0, 577.0] min_slope=-1.013 mean_slope=-0.673 drop_kdac=-2.026
gap_to_window=29.0 reason=excluded as noisy (multiple accepted transitions)
    row=22 col=16 transition=[508.0, 507.0] min_slope=-0.887 mean_slope=-0.882 drop_kdac=-1.753
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=23 col=16 transition=[587.0, 586.0] min_slope=-0.754 mean_slope=-0.754 drop_kdac=-1.507
gap_to_window=37.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=25 col=16 transition=[535.0, 535.0] min_slope=-0.093 mean_slope=-0.093 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=25 col=16 transition=[530.0, 529.0] min_slope=-0.106 mean_slope=-0.105 drop_kdac=-0.145
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=25 col=16 transition=[526.0, 518.0] min_slope=-0.188 mean_slope=-0.101 drop_kdac=-0.823
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=25 col=16 transition=[515.0, 515.0] min_slope=-0.110 mean_slope=-0.110 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
```

```
    row=25 col=16 transition=[512.0, 502.0] min_slope=-0.420 mean_slope=-0.232 drop_kdac=-2.379
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=31 col=16 transition=[521.0, 520.0] min_slope=-0.907 mean_slope=-0.877 drop_kdac=-1.810
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 17
  Noisy detectors excluded: [(4, 17), (14, 17), (23, 17), (25, 17), (29, 17)]
    row=4 col=17 noisy_transition=[1464.0, 1463.0] min_slope=-1.575 mean_slope=-1.397 drop_kdac=-3.075
    row=4 col=17 noisy_transition=[266.0, 264.0] min_slope=-3.047 mean_slope=-2.029 drop_kdac=-6.095
    row=14 col=17 noisy_transition=[1341.0, 1338.0] min_slope=-8.579 mean_slope=-4.650 drop_kdac=-18.542
    row=14 col=17 noisy_transition=[117.0, 109.0] min_slope=-2.373 mean_slope=-0.660 drop_kdac=-5.890
    row=23 col=17 noisy_transition=[1768.0, 1761.0] min_slope=-1.357 mean_slope=-0.631 drop_kdac=-4.944
    row=23 col=17 noisy_transition=[1201.0, 1200.0] min_slope=-1.341 mean_slope=-1.286 drop_kdac=-2.537
    row=23 col=17 noisy_transition=[1124.0, 1123.0] min_slope=-1.238 mean_slope=-1.233 drop_kdac=-2.477
    row=23 col=17 noisy_transition=[245.0, 240.0] min_slope=-8.733 mean_slope=-5.867 drop_kdac=-35.213
    row=23 col=17 noisy_transition=[222.0, 216.0] min_slope=-5.163 mean_slope=-3.044 drop_kdac=-21.306
    row=23 col=17 noisy_transition=[196.0, 192.0] min_slope=-3.445 mean_slope=-2.246 drop_kdac=-11.234
    row=23 col=17 noisy_transition=[179.0, 175.0] min_slope=-1.706 mean_slope=-0.967 drop_kdac=-4.997
    row=23 col=17 noisy_transition=[24.0, 23.0] min_slope=-1.331 mean_slope=-1.280 drop_kdac=-2.536
    row=25 col=17 noisy_transition=[1551.0, 1549.0] min_slope=-1.257 mean_slope=-0.888 drop_kdac=-2.514
    row=25 col=17 noisy_transition=[1305.0, 1303.0] min_slope=-1.246 mean_slope=-0.870 drop_kdac=-2.492
    row=25 col=17 noisy_transition=[1224.0, 1223.0] min_slope=-1.431 mean_slope=-1.384 drop_kdac=-2.776
    row=25 col=17 noisy_transition=[1133.0, 1131.0] min_slope=-1.253 mean_slope=-0.832 drop_kdac=-2.506
    row=25 col=17 noisy_transition=[989.0, 988.0] min_slope=-1.282 mean_slope=-1.258 drop_kdac=-2.398
    row=25 col=17 noisy_transition=[844.0, 839.0] min_slope=-1.397 mean_slope=-0.858 drop_kdac=-5.274
    row=25 col=17 noisy_transition=[571.0, 570.0] min_slope=-1.221 mean_slope=-1.215 drop_kdac=-2.387
    row=25 col=17 noisy_transition=[555.0, 554.0] min_slope=-1.500 mean_slope=-1.370 drop_kdac=-2.997
    row=25 col=17 noisy_transition=[449.0, 448.0] min_slope=-1.243 mean_slope=-1.223 drop_kdac=-2.405
    row=25 col=17 noisy_transition=[236.0, 234.0] min_slope=-3.746 mean_slope=-2.504 drop_kdac=-7.491
    row=25 col=17 noisy_transition=[22.0, 21.0] min_slope=-1.246 mean_slope=-1.240 drop_kdac=-2.463
    row=29 col=17 noisy_transition=[1776.0, 1774.0] min_slope=-1.347 mean_slope=-0.884 drop_kdac=-2.694
    row=29 col=17 noisy_transition=[1574.0, 1571.0] min_slope=-1.248 mean_slope=-0.578 drop_kdac=-2.344
    row=29 col=17 noisy_transition=[786.0, 784.0] min_slope=-1.238 mean_slope=-0.823 drop_kdac=-2.477
    row=29 col=17 noisy_transition=[751.0, 750.0] min_slope=-1.283 mean_slope=-1.271 drop_kdac=-2.506
    row=29 col=17 noisy_transition=[148.0, 146.0] min_slope=-1.253 mean_slope=-0.862 drop_kdac=-2.506
  Best bias = 876.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [796.0, 876.0]: [(8, 17)]
    row=8 col=17 transition=[797.0, 796.0] min_slope=-1.269 mean_slope=-1.246 drop_kdac=-2.483
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=23 col=17 transition=[24.0, 23.0] min_slope=-1.331 mean_slope=-1.280 drop_kdac=-2.536
gap_to_window=24.0 reason=excluded as noisy (multiple accepted transitions)
    row=25 col=17 transition=[22.0, 21.0] min_slope=-1.246 mean_slope=-1.240 drop_kdac=-2.463
```

```
gap_to_window=22.0 reason=excluded as noisy (multiple accepted transitions)

Column 18
  Noisy detectors excluded: [(7, 18), (13, 18), (18, 18), (19, 18)]
    row=7 col=18 noisy_transition=[1438.0, 1436.0] min_slope=-1.040 mean_slope=-0.688 drop_kdac=-2.081
    row=7 col=18 noisy_transition=[1301.0, 1300.0] min_slope=-1.079 mean_slope=-1.075 drop_kdac=-2.134
    row=7 col=18 noisy_transition=[1163.0, 1161.0] min_slope=-1.108 mean_slope=-0.735 drop_kdac=-2.217
    row=7 col=18 noisy_transition=[1020.0, 1019.0] min_slope=-1.052 mean_slope=-1.037 drop_kdac=-2.046
    row=7 col=18 noisy_transition=[880.0, 879.0] min_slope=-1.049 mean_slope=-1.011 drop_kdac=-1.983
    row=7 col=18 noisy_transition=[738.0, 737.0] min_slope=-1.034 mean_slope=-1.019 drop_kdac=-2.068
    row=7 col=18 noisy_transition=[591.0, 589.0] min_slope=-1.044 mean_slope=-0.690 drop_kdac=-2.089
    row=7 col=18 noisy_transition=[438.0, 436.0] min_slope=-1.006 mean_slope=-0.669 drop_kdac=-2.012
    row=7 col=18 noisy_transition=[319.0, 304.0] min_slope=-2.355 mean_slope=-0.452 drop_kdac=-7.239
    row=7 col=18 noisy_transition=[57.0, 56.0] min_slope=-1.002 mean_slope=-1.000 drop_kdac=-1.996
    row=7 col=18 noisy_transition=[22.0, 21.0] min_slope=-1.000 mean_slope=-0.999 drop_kdac=-2.000
    row=13 col=18 noisy_transition=[1488.0, 1487.0] min_slope=-1.011 mean_slope=-1.001 drop_kdac=-1.981
    row=13 col=18 noisy_transition=[1356.0, 1355.0] min_slope=-1.083 mean_slope=-1.055 drop_kdac=-2.162
    row=13 col=18 noisy_transition=[1219.0, 1218.0] min_slope=-1.067 mean_slope=-1.008 drop_kdac=-2.127
    row=13 col=18 noisy_transition=[1088.0, 1087.0] min_slope=-1.009 mean_slope=-1.000 drop_kdac=-1.976
    row=13 col=18 noisy_transition=[953.0, 951.0] min_slope=-1.006 mean_slope=-0.666 drop_kdac=-2.012
    row=13 col=18 noisy_transition=[819.0, 818.0] min_slope=-1.035 mean_slope=-1.025 drop_kdac=-2.026
    row=13 col=18 noisy_transition=[682.0, 681.0] min_slope=-1.038 mean_slope=-1.029 drop_kdac=-2.035
    row=13 col=18 noisy_transition=[543.0, 542.0] min_slope=-1.048 mean_slope=-1.039 drop_kdac=-2.053
    row=13 col=18 noisy_transition=[300.0, 279.0] min_slope=-1.895 mean_slope=-0.344 drop_kdac=-7.590
    row=13 col=18 noisy_transition=[182.0, 181.0] min_slope=-1.041 mean_slope=-1.030 drop_kdac=-2.074
    row=13 col=18 noisy_transition=[157.0, 156.0] min_slope=-1.072 mean_slope=-1.052 drop_kdac=-2.137
    row=13 col=18 noisy_transition=[132.0, 131.0] min_slope=-1.050 mean_slope=-1.040 drop_kdac=-2.095
    row=13 col=18 noisy_transition=[107.0, 106.0] min_slope=-1.055 mean_slope=-1.043 drop_kdac=-2.099
    row=13 col=18 noisy_transition=[82.0, 81.0] min_slope=-1.050 mean_slope=-1.038 drop_kdac=-2.090
    row=13 col=18 noisy_transition=[57.0, 56.0] min_slope=-1.039 mean_slope=-1.027 drop_kdac=-2.064
    row=13 col=18 noisy_transition=[32.0, 31.0] min_slope=-1.028 mean_slope=-1.016 drop_kdac=-2.045
    row=13 col=18 noisy_transition=[7.0, 6.0] min_slope=-1.031 mean_slope=-1.018 drop_kdac=-2.048
    row=18 col=18 noisy_transition=[1699.0, 1698.0] min_slope=-1.034 mean_slope=-1.023 drop_kdac=-2.022
    row=18 col=18 noisy_transition=[1533.0, 1532.0] min_slope=-1.052 mean_slope=-1.044 drop_kdac=-2.066
    row=18 col=18 noisy_transition=[1377.0, 1376.0] min_slope=-1.050 mean_slope=-1.041 drop_kdac=-2.060
    row=18 col=18 noisy_transition=[1231.0, 1230.0] min_slope=-1.059 mean_slope=-1.048 drop_kdac=-2.072
    row=18 col=18 noisy_transition=[1088.0, 1087.0] min_slope=-1.047 mean_slope=-1.037 drop_kdac=-2.052
    row=18 col=18 noisy_transition=[948.0, 947.0] min_slope=-1.041 mean_slope=-1.041 drop_kdac=-2.080
    row=18 col=18 noisy_transition=[809.0, 808.0] min_slope=-1.035 mean_slope=-1.027 drop_kdac=-2.033
    row=18 col=18 noisy_transition=[672.0, 671.0] min_slope=-1.048 mean_slope=-1.037 drop_kdac=-2.051
    row=18 col=18 noisy_transition=[535.0, 534.0] min_slope=-1.164 mean_slope=-1.105 drop_kdac=-2.324
    row=18 col=18 noisy_transition=[398.0, 397.0] min_slope=-1.051 mean_slope=-1.046 drop_kdac=-2.077
    row=18 col=18 noisy_transition=[260.0, 259.0] min_slope=-1.036 mean_slope=-1.028 drop_kdac=-2.036
    row=18 col=18 noisy_transition=[122.0, 121.0] min_slope=-1.030 mean_slope=-1.021 drop_kdac=-2.017
    row=19 col=18 noisy_transition=[243.0, 222.0] min_slope=-4.245 mean_slope=-0.531 drop_kdac=-11.786
```

```
    row=19 col=18 noisy_transition=[67.0, 66.0] min_slope=-1.006 mean_slope=-0.930 drop_kdac=-2.006
  Best bias = 320.0
  Max detectors in transition at best bias = 7
  Detectors at best bias in [240.0, 320.0]: [(15, 18), (21, 18), (22, 18), (28, 18), (29, 18), (30, 18), (31,
18)]
    row=15 col=18 transition=[264.0, 252.0] min_slope=-1.542 mean_slope=-0.409 drop_kdac=-5.326
    row=21 col=18 transition=[295.0, 269.0] min_slope=-3.967 mean_slope=-0.456 drop_kdac=-12.318
    row=22 col=18 transition=[316.0, 293.0] min_slope=-3.812 mean_slope=-0.465 drop_kdac=-11.147
    row=28 col=18 transition=[278.0, 258.0] min_slope=-4.475 mean_slope=-0.611 drop_kdac=-12.862
    row=29 col=18 transition=[310.0, 289.0] min_slope=-3.434 mean_slope=-0.465 drop_kdac=-10.165
    row=30 col=18 transition=[260.0, 240.0] min_slope=-4.963 mean_slope=-0.629 drop_kdac=-13.179
    row=31 col=18 transition=[318.0, 298.0] min_slope=-3.068 mean_slope=-0.457 drop_kdac=-9.603
  User bias = 325
  Clean detectors in transition in [245.0, 325.0]: [(15, 18), (21, 18), (22, 18), (28, 18), (29, 18), (31,
18)]
    row=15 col=18 transition=[264.0, 252.0] min_slope=-1.542 mean_slope=-0.409 drop_kdac=-5.326
    row=21 col=18 transition=[295.0, 269.0] min_slope=-3.967 mean_slope=-0.456 drop_kdac=-12.318
    row=22 col=18 transition=[316.0, 293.0] min_slope=-3.812 mean_slope=-0.465 drop_kdac=-11.147
    row=28 col=18 transition=[278.0, 258.0] min_slope=-4.475 mean_slope=-0.611 drop_kdac=-12.862
    row=29 col=18 transition=[310.0, 289.0] min_slope=-3.434 mean_slope=-0.465 drop_kdac=-10.165
    row=31 col=18 transition=[318.0, 298.0] min_slope=-3.068 mean_slope=-0.457 drop_kdac=-9.603
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=4 col=18 transition=[316.0, 315.0] min_slope=-0.845 mean_slope=-0.833 drop_kdac=-1.691
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=4 col=18 transition=[291.0, 290.0] min_slope=-0.834 mean_slope=-0.823 drop_kdac=-1.668
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=4 col=18 transition=[266.0, 265.0] min_slope=-0.874 mean_slope=-0.861 drop_kdac=-1.745
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=4 col=18 transition=[241.0, 240.0] min_slope=-0.882 mean_slope=-0.871 drop_kdac=-1.762
gap_to_window=5.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=4 col=18 transition=[216.0, 215.0] min_slope=-0.874 mean_slope=-0.861 drop_kdac=-1.744
gap_to_window=30.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=7 col=18 transition=[319.0, 304.0] min_slope=-2.355 mean_slope=-0.452 drop_kdac=-7.239
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=8 col=18 transition=[321.0, 304.0] min_slope=-0.502 mean_slope=-0.220 drop_kdac=-3.923
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=13 col=18 transition=[300.0, 279.0] min_slope=-1.895 mean_slope=-0.344 drop_kdac=-7.590
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=14 col=18 transition=[249.0, 232.0] min_slope=-1.416 mean_slope=-0.292 drop_kdac=-5.288
gap_to_window=13.0 reason=accepted transition outside user window
    row=18 col=18 transition=[260.0, 259.0] min_slope=-1.036 mean_slope=-1.028 drop_kdac=-2.036
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=19 col=18 transition=[243.0, 222.0] min_slope=-4.245 mean_slope=-0.531 drop_kdac=-11.786
gap_to_window=23.0 reason=excluded as noisy (multiple accepted transitions)
    row=20 col=18 transition=[365.0, 347.0] min_slope=-1.393 mean_slope=-0.326 drop_kdac=-6.153
```

```
gap_to_window=40.0 reason=accepted transition outside user window
    row=23 col=18 transition=[330.0, 311.0] min_slope=-0.955 mean_slope=-0.245 drop_kdac=-4.934
gap_to_window=5.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=24 col=18 transition=[284.0, 267.0] min_slope=-0.878 mean_slope=-0.222 drop_kdac=-4.188
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=30 col=18 transition=[260.0, 240.0] min_slope=-4.963 mean_slope=-0.629 drop_kdac=-13.179
gap_to_window=5.0 reason=accepted transition outside user window

Column 19
  Noisy detectors excluded: []
  Best bias = 300.0
  Max detectors in transition at best bias = 7
  Detectors at best bias in [220.0, 300.0]: [(1, 19), (5, 19), (8, 19), (20, 19), (21, 19), (24, 19), (31,
19)]
    row=1 col=19 transition=[257.0, 237.0] min_slope=-1.675 mean_slope=-0.320 drop_kdac=-6.735
    row=5 col=19 transition=[241.0, 220.0] min_slope=-1.719 mean_slope=-0.298 drop_kdac=-6.535
    row=8 col=19 transition=[266.0, 249.0] min_slope=-3.432 mean_slope=-0.538 drop_kdac=-9.767
    row=20 col=19 transition=[240.0, 221.0] min_slope=-2.466 mean_slope=-0.405 drop_kdac=-8.089
    row=21 col=19 transition=[275.0, 252.0] min_slope=-1.854 mean_slope=-0.306 drop_kdac=-7.349
    row=24 col=19 transition=[246.0, 226.0] min_slope=-1.282 mean_slope=-0.258 drop_kdac=-5.412
    row=31 col=19 transition=[260.0, 236.0] min_slope=-2.898 mean_slope=-0.379 drop_kdac=-9.504
  User bias = 400
  Clean detectors in transition in [320.0, 400.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=6 col=19 transition=[319.0, 292.0] min_slope=-2.721 mean_slope=-0.345 drop_kdac=-9.780
gap_to_window=28.0 reason=accepted transition outside user window
    row=9 col=19 transition=[387.0, 387.0] min_slope=-0.082 mean_slope=-0.082 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=9 col=19 transition=[384.0, 360.0] min_slope=-0.419 mean_slope=-0.146 drop_kdac=-3.651
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=10 col=19 transition=[371.0, 344.0] min_slope=-0.820 mean_slope=-0.187 drop_kdac=-5.295
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=17 col=19 transition=[351.0, 328.0] min_slope=-0.218 mean_slope=-0.122 drop_kdac=-2.772
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=18 col=19 transition=[368.0, 344.0] min_slope=-0.275 mean_slope=-0.125 drop_kdac=-3.143
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=19 col=19 transition=[303.0, 282.0] min_slope=-0.290 mean_slope=-0.135 drop_kdac=-2.779
gap_to_window=38.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=27 col=19 transition=[410.0, 384.0] min_slope=-0.297 mean_slope=-0.139 drop_kdac=-3.538
gap_to_window=10.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 20
  Noisy detectors excluded: []
  Best bias = 303.0
```

```
  Max detectors in transition at best bias = 4
  Detectors at best bias in [223.0, 303.0]: [(5, 20), (8, 20), (12, 20), (15, 20)]
    row=5 col=20 transition=[244.0, 225.0] min_slope=-3.748 mean_slope=-0.543 drop_kdac=-10.835
    row=8 col=20 transition=[282.0, 260.0] min_slope=-2.687 mean_slope=-0.379 drop_kdac=-8.798
    row=12 col=20 transition=[290.0, 267.0] min_slope=-2.629 mean_slope=-0.384 drop_kdac=-9.226
    row=15 col=20 transition=[242.0, 223.0] min_slope=-4.490 mean_slope=-0.607 drop_kdac=-12.111
  User bias = 350
  Clean detectors in transition in [270.0, 350.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=3 col=20 transition=[364.0, 339.0] min_slope=-0.963 mean_slope=-0.199 drop_kdac=-5.203
gap_to_window=14.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=4 col=20 transition=[312.0, 289.0] min_slope=-0.875 mean_slope=-0.211 drop_kdac=-5.135
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=7 col=20 transition=[261.0, 240.0] min_slope=-0.409 mean_slope=-0.165 drop_kdac=-3.564
gap_to_window=30.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=8 col=20 transition=[282.0, 260.0] min_slope=-2.687 mean_slope=-0.379 drop_kdac=-8.798
gap_to_window=10.0 reason=accepted transition outside user window
    row=11 col=20 transition=[300.0, 279.0] min_slope=-0.496 mean_slope=-0.172 drop_kdac=-3.796
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=12 col=20 transition=[290.0, 267.0] min_slope=-2.629 mean_slope=-0.384 drop_kdac=-9.226
gap_to_window=3.0 reason=accepted transition outside user window
    row=16 col=20 transition=[279.0, 259.0] min_slope=-0.472 mean_slope=-0.161 drop_kdac=-3.490
gap_to_window=11.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=17 col=20 transition=[245.0, 231.0] min_slope=-0.257 mean_slope=-0.145 drop_kdac=-2.022
gap_to_window=39.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=19 col=20 transition=[279.0, 262.0] min_slope=-0.301 mean_slope=-0.147 drop_kdac=-2.598
gap_to_window=8.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=25 col=20 transition=[294.0, 293.0] min_slope=-0.688 mean_slope=-0.653 drop_kdac=-1.393
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=28 col=20 transition=[379.0, 365.0] min_slope=-0.305 mean_slope=-0.142 drop_kdac=-1.911
gap_to_window=29.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 21
  Noisy detectors excluded: []
  Best bias = 259.0
  Max detectors in transition at best bias = 3
  Detectors at best bias in [179.0, 259.0]: [(12, 21), (19, 21), (20, 21)]
    row=12 col=21 transition=[200.0, 179.0] min_slope=-2.603 mean_slope=-0.403 drop_kdac=-8.888
    row=19 col=21 transition=[240.0, 217.0] min_slope=-3.268 mean_slope=-0.438 drop_kdac=-10.497
    row=20 col=21 transition=[211.0, 191.0] min_slope=-4.428 mean_slope=-0.595 drop_kdac=-12.458
  User bias = 375
  Clean detectors in transition in [295.0, 375.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=6 col=21 transition=[294.0, 271.0] min_slope=-1.077 mean_slope=-0.209 drop_kdac=-5.032
gap_to_window=24.0 reason=accepted transition outside user window
```

```
    row=16 col=21 transition=[313.0, 290.0] min_slope=-0.610 mean_slope=-0.196 drop_kdac=-4.690
gap_to_window=5.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=17 col=21 transition=[316.0, 291.0] min_slope=-0.444 mean_slope=-0.159 drop_kdac=-4.122
gap_to_window=4.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=18 col=21 transition=[395.0, 375.0] min_slope=-0.999 mean_slope=-0.195 drop_kdac=-4.079
gap_to_window=20.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=21 col=21 transition=[317.0, 293.0] min_slope=-0.925 mean_slope=-0.195 drop_kdac=-4.903
gap_to_window=2.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=23 col=21 transition=[327.0, 304.0] min_slope=-0.966 mean_slope=-0.219 drop_kdac=-5.270
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=24 col=21 transition=[415.0, 390.0] min_slope=-1.155 mean_slope=-0.195 drop_kdac=-5.062
gap_to_window=40.0 reason=accepted transition outside user window
    row=25 col=21 transition=[320.0, 296.0] min_slope=-0.595 mean_slope=-0.170 drop_kdac=-4.272
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=27 col=21 transition=[335.0, 311.0] min_slope=-0.772 mean_slope=-0.192 drop_kdac=-4.817
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=21 transition=[306.0, 285.0] min_slope=-0.296 mean_slope=-0.150 drop_kdac=-3.167
gap_to_window=10.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=31 col=21 transition=[359.0, 335.0] min_slope=-0.256 mean_slope=-0.129 drop_kdac=-3.246
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 22
  Noisy detectors excluded: [(5, 22)]
    row=5 col=22 noisy_transition=[893.0, 892.0] min_slope=-1.084 mean_slope=-1.031 drop_kdac=-2.167
    row=5 col=22 noisy_transition=[720.0, 716.0] min_slope=-2.015 mean_slope=-0.849 drop_kdac=-4.338
    row=5 col=22 noisy_transition=[187.0, 186.0] min_slope=-1.061 mean_slope=-1.014 drop_kdac=-2.116
  Best bias = 997.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [917.0, 997.0]: [(11, 22), (13, 22)]
    row=11 col=22 transition=[921.0, 917.0] min_slope=-1.131 mean_slope=-0.513 drop_kdac=-2.511
    row=13 col=22 transition=[921.0, 917.0] min_slope=-1.835 mean_slope=-0.809 drop_kdac=-3.984
  User bias = 900
  Clean detectors in transition in [820.0, 900.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=5 col=22 transition=[893.0, 892.0] min_slope=-1.084 mean_slope=-1.031 drop_kdac=-2.167
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=7 col=22 transition=[795.0, 789.0] min_slope=-1.656 mean_slope=-0.574 drop_kdac=-3.984
gap_to_window=31.0 reason=accepted transition outside user window
    row=8 col=22 transition=[916.0, 915.0] min_slope=-0.833 mean_slope=-0.787 drop_kdac=-1.667
gap_to_window=16.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=9 col=22 transition=[891.0, 888.0] min_slope=-0.323 mean_slope=-0.173 drop_kdac=-0.434
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=10 col=22 transition=[920.0, 919.0] min_slope=-0.794 mean_slope=-0.784 drop_kdac=-1.549
gap_to_window=20.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=11 col=22 transition=[921.0, 917.0] min_slope=-1.131 mean_slope=-0.513 drop_kdac=-2.511
```

```
gap_to_window=21.0 reason=accepted transition outside user window
    row=13 col=22 transition=[921.0, 917.0] min_slope=-1.835 mean_slope=-0.809 drop_kdac=-3.984
gap_to_window=21.0 reason=accepted transition outside user window
    row=24 col=22 transition=[849.0, 848.0] min_slope=-0.606 mean_slope=-0.597 drop_kdac=-1.184
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=25 col=22 transition=[788.0, 786.0] min_slope=-0.571 mean_slope=-0.396 drop_kdac=-1.143
gap_to_window=34.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=27 col=22 transition=[889.0, 887.0] min_slope=-0.616 mean_slope=-0.406 drop_kdac=-1.232
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 23
  Noisy detectors excluded: []
  Best bias = 457.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [377.0, 457.0]: [(27, 23)]
    row=27 col=23 transition=[400.0, 377.0] min_slope=-1.008 mean_slope=-0.186 drop_kdac=-4.466
  User bias = 450
  Clean detectors in transition in [370.0, 450.0]: [(27, 23)]
    row=27 col=23 transition=[400.0, 377.0] min_slope=-1.008 mean_slope=-0.186 drop_kdac=-4.466
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=13 col=23 transition=[366.0, 364.0] min_slope=-0.644 mean_slope=-0.454 drop_kdac=-1.287
gap_to_window=6.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=13 col=23 transition=[359.0, 341.0] min_slope=-0.188 mean_slope=-0.118 drop_kdac=-2.109
gap_to_window=29.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=15 col=23 transition=[365.0, 341.0] min_slope=-0.728 mean_slope=-0.164 drop_kdac=-4.120
gap_to_window=29.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=17 col=23 transition=[421.0, 398.0] min_slope=-0.394 mean_slope=-0.127 drop_kdac=-3.053
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=21 col=23 transition=[425.0, 402.0] min_slope=-0.418 mean_slope=-0.133 drop_kdac=-3.215
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 24
  Noisy detectors excluded: []
  Best bias = 329.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [249.0, 329.0]: [(18, 24)]
    row=18 col=24 transition=[266.0, 249.0] min_slope=-4.197 mean_slope=-0.606 drop_kdac=-11.143
  User bias = 250
  Clean detectors in transition in [170.0, 250.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=18 col=24 transition=[266.0, 249.0] min_slope=-4.197 mean_slope=-0.606 drop_kdac=-11.143
gap_to_window=16.0 reason=accepted transition outside user window
    row=26 col=24 transition=[160.0, 145.0] min_slope=-2.430 mean_slope=-0.470 drop_kdac=-7.519
gap_to_window=25.0 reason=accepted transition outside user window
```

```
Column 25
  Noisy detectors excluded: [(16, 25)]
    row=16 col=25 noisy_transition=[1006.0, 1004.0] min_slope=-1.503 mean_slope=-0.895 drop_kdac=-3.005
    row=16 col=25 noisy_transition=[338.0, 333.0] min_slope=-1.012 mean_slope=-0.250 drop_kdac=-1.470
    row=16 col=25 noisy_transition=[330.0, 324.0] min_slope=-1.600 mean_slope=-0.673 drop_kdac=-4.747
  Best bias = 371.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [291.0, 371.0]: [(12, 25)]
    row=12 col=25 transition=[323.0, 291.0] min_slope=-1.055 mean_slope=-0.177 drop_kdac=-5.834
  User bias = 400
  Clean detectors in transition in [320.0, 400.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=12 col=25 transition=[323.0, 291.0] min_slope=-1.055 mean_slope=-0.177 drop_kdac=-5.834
gap_to_window=29.0 reason=accepted transition outside user window
    row=16 col=25 transition=[338.0, 333.0] min_slope=-1.012 mean_slope=-0.250 drop_kdac=-1.470
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=16 col=25 transition=[330.0, 324.0] min_slope=-1.600 mean_slope=-0.673 drop_kdac=-4.747
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=19 col=25 transition=[367.0, 351.0] min_slope=-0.606 mean_slope=-0.159 drop_kdac=-2.729
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=20 col=25 transition=[438.0, 437.0] min_slope=-0.227 mean_slope=-0.224 drop_kdac=-0.421
gap_to_window=38.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=20 col=25 transition=[403.0, 402.0] min_slope=-0.088 mean_slope=-0.087 drop_kdac=-0.172
gap_to_window=3.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 26
  Noisy detectors excluded: [(11, 26), (14, 26)]
    row=11 col=26 noisy_transition=[468.0, 466.0] min_slope=-1.209 mean_slope=-0.828 drop_kdac=-2.418
    row=11 col=26 noisy_transition=[456.0, 454.0] min_slope=-1.359 mean_slope=-0.843 drop_kdac=-2.718
    row=11 col=26 noisy_transition=[433.0, 429.0] min_slope=-1.234 mean_slope=-0.467 drop_kdac=-2.180
    row=11 col=26 noisy_transition=[366.0, 360.0] min_slope=-1.379 mean_slope=-0.678 drop_kdac=-4.905
    row=11 col=26 noisy_transition=[299.0, 296.0] min_slope=-1.013 mean_slope=-0.617 drop_kdac=-2.464
    row=11 col=26 noisy_transition=[293.0, 284.0] min_slope=-11.627 mean_slope=-4.912 drop_kdac=-42.367
    row=11 col=26 noisy_transition=[281.0, 276.0] min_slope=-15.792 mean_slope=-5.086 drop_kdac=-20.607
    row=11 col=26 noisy_transition=[175.0, 172.0] min_slope=-4.331 mean_slope=-2.569 drop_kdac=-10.298
    row=14 col=26 noisy_transition=[911.0, 891.0] min_slope=-4.026 mean_slope=-2.291 drop_kdac=-48.183
    row=14 col=26 noisy_transition=[872.0, 869.0] min_slope=-1.282 mean_slope=-0.725 drop_kdac=-2.413
    row=14 col=26 noisy_transition=[856.0, 847.0] min_slope=-2.331 mean_slope=-0.893 drop_kdac=-8.188
    row=14 col=26 noisy_transition=[841.0, 833.0] min_slope=-7.981 mean_slope=-4.531 drop_kdac=-40.603
    row=14 col=26 noisy_transition=[818.0, 809.0] min_slope=-1.230 mean_slope=-0.427 drop_kdac=-4.022
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 350
  Clean detectors in transition in [270.0, 350.0]: []
```

```
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=1 col=26 transition=[274.0, 255.0] min_slope=-0.800 mean_slope=-0.276 drop_kdac=-5.537
gap_to_window=15.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=3 col=26 transition=[312.0, 312.0] min_slope=-0.082 mean_slope=-0.082 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=3 col=26 transition=[306.0, 287.0] min_slope=-0.496 mean_slope=-0.160 drop_kdac=-3.200
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=7 col=26 transition=[288.0, 268.0] min_slope=-0.751 mean_slope=-0.189 drop_kdac=-3.973
gap_to_window=2.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=11 col=26 transition=[366.0, 360.0] min_slope=-1.379 mean_slope=-0.678 drop_kdac=-4.905
gap_to_window=16.0 reason=excluded as noisy (multiple accepted transitions)
    row=11 col=26 transition=[299.0, 296.0] min_slope=-1.013 mean_slope=-0.617 drop_kdac=-2.464
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=11 col=26 transition=[293.0, 284.0] min_slope=-11.627 mean_slope=-4.912 drop_kdac=-42.367
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=11 col=26 transition=[281.0, 276.0] min_slope=-15.792 mean_slope=-5.086 drop_kdac=-20.607
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=12 col=26 transition=[292.0, 292.0] min_slope=-0.104 mean_slope=-0.104 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=13 col=26 transition=[366.0, 365.0] min_slope=-0.133 mean_slope=-0.109 drop_kdac=-0.158
gap_to_window=16.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=13 col=26 transition=[300.0, 299.0] min_slope=-0.113 mean_slope=-0.107 drop_kdac=-0.229
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 27
  Noisy detectors excluded: [(29, 27)]
    row=29 col=27 noisy_transition=[1738.0, 1736.0] min_slope=-1.357 mean_slope=-0.875 drop_kdac=-2.713
    row=29 col=27 noisy_transition=[1034.0, 1026.0] min_slope=-1.084 mean_slope=-0.247 drop_kdac=-2.198
    row=29 col=27 noisy_transition=[805.0, 796.0] min_slope=-2.634 mean_slope=-1.502 drop_kdac=-14.584
    row=29 col=27 noisy_transition=[776.0, 775.0] min_slope=-1.297 mean_slope=-1.270 drop_kdac=-2.588
    row=29 col=27 noisy_transition=[768.0, 767.0] min_slope=-1.485 mean_slope=-1.467 drop_kdac=-2.862
    row=29 col=27 noisy_transition=[764.0, 762.0] min_slope=-1.244 mean_slope=-0.833 drop_kdac=-2.489
    row=29 col=27 noisy_transition=[639.0, 638.0] min_slope=-1.078 mean_slope=-0.601 drop_kdac=0.132
    row=29 col=27 noisy_transition=[497.0, 493.0] min_slope=-1.462 mean_slope=-0.533 drop_kdac=-2.945
    row=29 col=27 noisy_transition=[428.0, 420.0] min_slope=-14.155 mean_slope=-5.106 drop_kdac=-45.873
    row=29 col=27 noisy_transition=[353.0, 351.0] min_slope=-1.555 mean_slope=-0.950 drop_kdac=-3.110
  Best bias = 517.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [437.0, 517.0]: [(9, 27), (28, 27)]
    row=9 col=27 transition=[497.0, 493.0] min_slope=-1.123 mean_slope=-0.516 drop_kdac=-2.490
    row=28 col=27 transition=[443.0, 437.0] min_slope=-2.255 mean_slope=-0.947 drop_kdac=-6.559
  User bias = 750
  Clean detectors in transition in [670.0, 750.0]: []
```

```
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=10 col=27 transition=[647.0, 642.0] min_slope=-0.533 mean_slope=-0.231 drop_kdac=-1.293
gap_to_window=28.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=27 transition=[776.0, 775.0] min_slope=-1.297 mean_slope=-1.270 drop_kdac=-2.588
gap_to_window=26.0 reason=excluded as noisy (multiple accepted transitions)
    row=29 col=27 transition=[768.0, 767.0] min_slope=-1.485 mean_slope=-1.467 drop_kdac=-2.862
gap_to_window=18.0 reason=excluded as noisy (multiple accepted transitions)
    row=29 col=27 transition=[764.0, 762.0] min_slope=-1.244 mean_slope=-0.833 drop_kdac=-2.489
gap_to_window=14.0 reason=excluded as noisy (multiple accepted transitions)
    row=29 col=27 transition=[639.0, 638.0] min_slope=-1.078 mean_slope=-0.601 drop_kdac=0.132
gap_to_window=32.0 reason=excluded as noisy (multiple accepted transitions)

Column 28
  Noisy detectors excluded: [(10, 28)]
    row=10 col=28 noisy_transition=[1077.0, 1073.0] min_slope=-1.287 mean_slope=-0.522 drop_kdac=-2.621
    row=10 col=28 noisy_transition=[1049.0, 1047.0] min_slope=-1.249 mean_slope=-0.828 drop_kdac=-2.499
    row=10 col=28 noisy_transition=[1008.0, 1007.0] min_slope=-1.314 mean_slope=-1.287 drop_kdac=-2.642
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 950
  Clean detectors in transition in [870.0, 950.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 29
  Noisy detectors excluded: []
  Best bias = 636.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [556.0, 636.0]: [(3, 29)]
    row=3 col=29 transition=[565.0, 556.0] min_slope=-1.096 mean_slope=-0.306 drop_kdac=-3.059
  User bias = 900
  Clean detectors in transition in [820.0, 900.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=17 col=29 transition=[930.0, 929.0] min_slope=-0.212 mean_slope=-0.190 drop_kdac=-0.413
gap_to_window=30.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=17 col=29 transition=[900.0, 899.0] min_slope=-0.168 mean_slope=-0.137 drop_kdac=-0.301
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=17 col=29 transition=[871.0, 870.0] min_slope=-0.151 mean_slope=-0.118 drop_kdac=-0.243
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=17 col=29 transition=[842.0, 842.0] min_slope=-0.141 mean_slope=-0.141 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=17 col=29 transition=[814.0, 813.0] min_slope=-0.169 mean_slope=-0.130 drop_kdac=-0.255
gap_to_window=7.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
```

```
    row=17 col=29 transition=[786.0, 785.0] min_slope=-0.185 mean_slope=-0.158 drop_kdac=-0.333
gap_to_window=35.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[925.0, 924.0] min_slope=-0.175 mean_slope=-0.147 drop_kdac=-0.319
gap_to_window=25.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[896.0, 895.0] min_slope=-0.159 mean_slope=-0.131 drop_kdac=-0.285
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[867.0, 866.0] min_slope=-0.147 mean_slope=-0.117 drop_kdac=-0.256
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[839.0, 838.0] min_slope=-0.189 mean_slope=-0.162 drop_kdac=-0.353
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[810.0, 809.0] min_slope=-0.171 mean_slope=-0.144 drop_kdac=-0.311
gap_to_window=11.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[781.0, 780.0] min_slope=-0.156 mean_slope=-0.127 drop_kdac=-0.277
gap_to_window=40.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=32 col=29 transition=[926.0, 925.0] min_slope=-0.147 mean_slope=-0.120 drop_kdac=-0.267
gap_to_window=26.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=32 col=29 transition=[897.0, 897.0] min_slope=-0.133 mean_slope=-0.133 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=29 transition=[868.0, 868.0] min_slope=-0.116 mean_slope=-0.116 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=29 transition=[839.0, 839.0] min_slope=-0.099 mean_slope=-0.099 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=32 col=29 transition=[811.0, 810.0] min_slope=-0.135 mean_slope=-0.108 drop_kdac=-0.240
gap_to_window=10.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=32 col=29 transition=[782.0, 782.0] min_slope=-0.119 mean_slope=-0.119 drop_kdac=0.000
gap_to_window=38.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule

Column 30
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 31
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
```

```
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Saved PDF:
/home/timestudent/anie/load_curves_all/specB/best_bias_20260129a_specB_iv_skyK_374mk_iv_all_data.pdf
```